



TECHNICAL PROPOSAL — RFP NVIDIA-REG-RD-2026-1106

SECURE ENTERPRISE NETWORK

NVIDIA Regional R&D and Production Hub

CLIENT

NVIDIA
CORPORATION

PREPARED BY

SHARP NETWORKS — BRUSSELS,
BE

ISSUED

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EXECUTIVE SUMMARY

This document constitutes the response of 锐| Sharp Networks (referred to henceforth as "Sharp Networks", "us", "we", "our team", "our company" or "the company") to the demands published by NVIDIA Corporation (referred to henceforth as "NVIDIA", "you", or "your company") in a request for proposal (referred to henceforth as "the request" or "the RFP") issued on June 11th, 2026 (See NVIDIA-REG-RD-2026-1106), along with the clarifications issued by NVIDIA's Jay Park.

We are honored to submit this bid for NVIDIA's consideration, and it is our hope to be selected to collaborate with your company for this project and beyond, establishing a mutually beneficial business relationship based in trust and excellence in deliverables.

It is our opinion that NVIDIA's request, particularly as it pertains to its core technical demands, falls within the area of expertise of our team, and we are confident that our bid has the potential to exceed NVIDIA's expectations. As will transpire further, the request left us sufficient headroom to design a tailored system which implements best-in-class security practices without compromising on usability or flexibility. The request's mention of the hub's inclusion of AI training activities in particular alerted our team to the critical need for easy no-refactor scalability, compute prioritization, and robust uptime through redundancy.

In this proposal, we seek to answer every question you might have: *what*, *why*, *how*, and *how much*? We put our entire project team at your disposition for any and all questions, seven days a week from today until your decision is made or until the project is complete.

This document is aimed to be best reviewed by more technically inclined NVIDIA personnel, but sections related to our Design Philosophy, Budget Overview, and Future Considerations have been written to be fully approachable regardless of technical inclination.

RFQ COMPLIANCE MATRIX

The table below maps every requirement expressed in the RFQ to the section(s) of this proposal where it is designed, documented, and costed. No requirement is left unaddressed.

#	RFQ REQUIREMENT	WHERE ADDRESSED IN THIS PROPOSAL
1	48 workstations	Requirements Analysis (Network Sectors: 5+8+10+10+10+5 = 48); Budget Overview — Workstations & Peripherals (38 + 10 units)
2	Multiple departmental VLANs	Addressing and VLANs (14 VLANs incl. per-department segmentation); Access Control Matrix
3	Hierarchical enterprise architecture	Network Architecture — Logical Topology (redundant collapsed-core, two-tier hierarchical model); Physical Topology
4	High-performance production workstations	Requirements Analysis (High-End Production Hardware); Budget Overview (10 × MSI i7-14700F / RTX 5070 workstations); Compute Unit (4 × NVIDIA H100)
5	DNS	Services — DNS (split-horizon, internal server 10.20.70.10); Configuration Reference
6	DHCP	Services — DHCP (centralized server, per-VLAN pools, ip helper-address relay); Addressing Table
7	AAA authentication	Security Architecture — AAA and RADIUS (server 10.20.70.11, accounting via syslog); appendix RADIUS tables
8	HTTP services	Services — HTTPS (public-facing web server on DMZ, 10.20.60.10, static NAT)
9	FTP services	Services — FTP (centralized storage server 10.20.80.10, application-level access control)
10	DMZ implementation	Security Architecture — DMZ Design (dedicated VLAN 60, core-switch ACL isolation + ASA NAT/ACL policies)
11	ACL-based internal security	Security Architecture — ACL Strategy; Access Control Matrix; full per-VLAN ACL rules in appendix
12	Hardware firewall	Security Architecture — Perimeter Security (Cisco ASA, stateful inspection); Budget Overview — Network Devices
13	Professional branding	Sharp Networks branded proposal (cover, structure, diagrams)
14	Bill of Materials (BOM)	Budget Overview — itemized CAPEX (Workstations, Network Devices, Cabling, Server Room, Compute, Labor)
15	Budget not exceeding EUR 300,000	Budget Overview — Total CAPEX EUR 299,028.63, within the EUR 300,000 envelope

REQUIREMENTS ANALYSIS

3.1 NVIDIA'S REQUIREMENTS

NVIDIA's request (including the clarifications provided by Jay following the issuance of the RFP) included elements in the form of **devices**, **practices**, **services**, and a specific request of **network sectors** for six separate departments with differing sizes and needs from a workload and network privilege point of view. Below, you may find every requirement expressed in NVIDIA's request as understood by our team.

It is our pleasure to confirm that every single item mentioned by NVIDIA has been successfully integrated and designed coherently in our architecture to provide state-of-the-art security without significant compromises on bandwidth, flexibility, or usability.

- / **High-End Production Hardware** — Through clarification after issuance of the RFQ, this appeared to be the centerpiece and main value driver within the hub. As such, both as a need and as a consideration, a centralized GPU server was given the utmost consideration in budget and design. Further discussions with NVIDIA's Jay Park highlighted that the main compute of the hub should be centralized, and that four business printers ought to be provided.
- / **Dedicated Edge Firewall** — A physical device serving firewall needs for the entire local area network of the hub to prevent undesired traffic from ever reaching into the network.
- / **Security Services** — Specifically: **DNS**, **DHCP** and **AAA** services. Our architecture successfully implements a split-horizon DNS concept. DHCP is the default addressing method for every device within the network using predetermined pools. An AAA server is provided, with the network configured so as to let NVIDIA's IT department oversee network use and, if need be, catch malicious behavior in time; accounting in particular is realized through system logging on the firewall and on the core of the network (see below).
- / **DMZ** — The DMZ concept, implemented through VLANs and access control lists as specified in NVIDIA's requirements.
- / **Storage** — Though no description is provided, we assume this refers to a centrally accessible FTP server shared by the various employees working within the hub.
- / **Communication** — Secure communication between departments and servers. Our understanding is that communication between network sectors should be possible only via internet traffic (unless the ACL is modified by NVIDIA's IT department); other communication protocols between the various machines are restricted to least-privilege based on our zero-trust policy principles (see below).
- / **Network Sectors** — Management: 5 workstations · Study: 8 workstations · Production: 10 workstations (high-end) · Support A: 10 workstations · Support B: 10 workstations · IT Department: 5 workstations

3.2 OUR ANALYSIS

The key decision when designing this network was to merge OSI layers 2 and 3 into a collapsed-core extended star architecture. This means that two interconnected multi-layer switches perform the bulk of inter-VLAN communication. Cisco's proprietary HSRPv2 protocol grants us the ability to automatically fail over to the second switch in case the first one fails. This happens smoothly since both core switches are connected to every access switch and to the physical firewall. This decision also allows NVIDIA's IT personnel to restart and reconfigure either switch without making the network experience downtime.

Based on NVIDIA's requirements, we opted to maximize budget allocation to the centralized GPU compute node to reduce overhead and maximize the potential for return on investment, without sacrificing security, redundancy, or throughput guarantees.

The most pragmatic way to implement the network segmentation is via VLANs and access control lists separating the network flow of the various departments. This requires the development and implementation of a comprehensive access control matrix, which is then translated and implemented in the form of access control lists on the core switches.

NVIDIA's RFQ specifically required a DMZ concept implemented through VLANs. This means the DMZ server resides within the internal network on its own VLAN, with isolation enforced through ACL rules on the core switches and NAT/ACL policies on the ASA firewall. While this offers a thinner protection layer compared to a physically separate DMZ segment (since the DMZ shares the same switching infrastructure), coherent implementation of access control ensures the safety of this approach.

DESIGN PHILOSOPHY

Our philosophy for the design of this network follows a soft hierarchy of priorities:

1 SECURITY

We believe that the security apparatus of this network should come as the first priority, as it represents the single most important factor within our control to ensure the safety of your company's assets. While redundancy prevents downtime, proper defensive measures protect against threat actors who could steal or damage your company's assets in irreversible ways. We are aware that in the particular case of AI training, threat actors stealing model weights, polluting training data, or otherwise interfering with the pipeline can render months or years of work useless in an instant. This led to the implementation of least-privilege access by default to all devices and resources on the network, applying to every single device and VLAN in our architecture.

2 COMPUTE

We understand that the centralized compute unit is the single largest value generator and key piece of the network. This means that at all times, consideration was given to maximizing budget allocation for the central computing server and orienting workflow processes around the production units. Indeed, in the final budget, the centralized compute server represents approximately 55% of the total bill of materials.

3 SCALABILITY

Due to the novelty of this hub for NVIDIA, and in conjunction with the possibilities brought by the AI market today, we designed our system to handle 3x growth of all departments without major refactoring. The switches were sized appropriately, and strategic use of static and dynamic addressing ensures that the network remains plug-and-play, decreasing the workload on NVIDIA's IT department until the 3x cap is reached.

4 REDUNDANCY

We establish a system of passive and active redundancies to minimize single points of failure throughout the system. Some remain, but this is inherent to the budget constraints. Our choice to build the network around a duo of core switches and to double connected ports allows for full-duplex capabilities between the core and department-specific switches and, in case of port failure, maintains basic connectivity before a technician can fix or replace the faulty cable or device.

NETWORK ARCHITECTURE

This section covers the physical and logical topologies of our network, their designs, features, strengths and weaknesses, and should help future IT personnel develop and maintain the network.

5.1 LOGICAL TOPOLOGY

As stated prior, our network is built as a collapsed-core topology centered around a duo of multilayer (or core) switches. The main reason is that the scale of this network (48 workstations, scale-able to 144) does not justify a dedicated distribution layer, which would have caused excessive overhead in setup, maintenance, and operations.

From a logical perspective, these cores serve to permit or deny traffic between the various VLANs (see below). The core switches are configured to be agnostic with regard to physical ports and only examine the VLAN from which packets originate and towards which they are headed. Moving the identification of traffic from physical ports to VLANs provides more flexibility and an extra layer of protection against human error should a technician accidentally re-wire the core switch.

Each department, along with the service servers, is assigned a regular Layer 2 switch. Each of these switches is connected to both central core switches.

5.2 PHYSICAL TOPOLOGY

Two major constraints restrict our margin of maneuver regarding the physical topology: first, the office layout, which requires IT and Production departments to be on the ground floor (every other department on the first floor), and second, the centralization of networking devices in order to make upkeep easier and increase security by not leaving networking devices exposed within the office. While this implies longer cable runs, we consider this a minor inconvenience in exchange for increased security and manageability.

With both of these constraints in mind, we dedicate a space on the ground floor as a standalone server room. It includes all of the major networking components including servers, core and intermediary switches, along with the firewall. This also takes advantage of the AC unit required for the compute server.

Our physical topology requires extensive cabling, especially due to the use of two core switches working in tandem for redundancy. We use patch panels to ensure all cabling does not need to be redone in case one of the core switches must be swapped out.

Cable runs are pulled from the main server room to the upper floor; this is discussed more in the labor and installation part of the document.

Guest WiFi is installed strategically to optimize coverage office-wide.

5.3 THROUGHPUT PLANNING

The Catalyst 3650-24PS core switches provide a switching capacity of 176 Gbps and a forwarding rate of approximately 130 million packets per second, which exceeds the aggregate demand of this network by a wide margin. Each workstation connects to its access switch at 1 Gbps, and trunk links between access switches and the core operate at 1 Gbps. The EtherChannel between the two core switches aggregates two physical links for increased inter-core bandwidth and redundancy.

For capacity estimation, we assume an average sustained bandwidth of 5 Mbps per workstation during normal office operations, with production workstations capable of bursting significantly higher during data transfers to the compute server. At current scale (48 workstations), aggregate internal traffic is estimated at approximately 240 Mbps sustained, well within the capacity of the core switching fabric. At 3x scale (144 workstations), aggregate traffic would reach approximately 720 Mbps, which remains within the capacity of the existing core infrastructure without hardware replacement.

The primary throughput constraint is the ASA firewall, which handles all internet-bound and DMZ traffic. The ASA 5506-X provides approximately 750 Mbps of stateful inspection throughput. Since only internet-bound traffic traverses the ASA (internal inter-VLAN traffic is routed directly by the core switches), this capacity is sufficient for the current deployment and remains adequate at 3x scale, assuming typical office internet usage patterns. Should NVIDIA's internet bandwidth requirements grow beyond this threshold, the ASA can be upgraded independently without changes to the internal network architecture.

HARDWARE NOTE – The throughput figures above are based on the Cisco Catalyst 3650-24PS and ASA 5506-X platforms used in the Packet Tracer simulation and constitute a conservative capacity baseline. The production hardware specified in the Bill of Materials (Catalyst 9300-24T-E core switches, Firepower 1010 firewall) meets or exceeds these performance characteristics.

ADDRESSING AND VLANS

6.1 ADDRESSING

We base our addressing scheme on a series of VLANs which are defined below. We chose to exclusively work with IPv4 for ease of use, as security concerns associated with IPv4 versus IPv6 are otherwise minimal for this scope and mitigated by sane ACL rules. We use the internal network of 10.0.0.0 to ease readability of addresses for NVIDIA's IT personnel in the future. This also grants comfortable headroom for scalability.

Department host devices, at the exception of printers, are assigned IP addresses automatically via DHCP. This grants network flexibility for plugging and unplugging new devices without requiring IT intervention. Servers, however, are assigned static IP addresses, which is standard practice for a network of this scale.

6.2 SCHEME

Our addressing scheme works based on established VLANs which containerize traffic between the various departments and devices. They are as follows:

VLAN	NAME	PURPOSE
2	TRANSIT	Carries routed traffic between core switches and the ASA.
10	MGMT_SECRETARIAT	Administrative and managerial office.
20	STUDY	Research & Development (R&D), Engineering, and Lab teams.
30	PRODUCTION	Core production business.
40	SUPPORT_A	Customer service, technical support, or helpdesk.
45	SUPPORT_B	Customer service, technical support, or helpdesk.
50	IT_DEPT	IT and systems administration.
60	DMZ	Semi-public security perimeter.
70	INT_SERVERS	Private core infrastructure layer.
75	SRV_COMPUTE	Dedicated to GPU compute AI workloads.
80	STORAGE	High-speed data and backup FTP server.
99	NET_MGMT	Isolated hardware management network.
110	GUEST_WIFI	Guest wireless internet access.
999	BLACKHOLE	Security isolation trap (no SVI, no routing).

6.3 NOTES

The scheme logic works such that all internal user traffic is routed from a host device to the access switch, which carries it over to one of the core switches. The core switch checks whether the packet is authorized to transit via its ACL, then either denies it or forwards it to the appropriate device.

VLAN 2 exclusively carries routed traffic between the core switches and the ASA. All outbound internet traffic, inbound connections to the DMZ, and default route forwarding traverse this link.

VLAN 999 has no SVI and no IP addressing. It is applied to all unused ports on all devices as a security precaution to prevent undesirable traffic from exploiting idle ports.

6.4 ADDRESSING TABLE

Each VLAN is given a full byte of addressing space (the last octet of the IPv4 address). While this grants extra unused addressing space, it makes it significantly easier for IT personnel to identify networks and devices.

Every address within the network takes the form of 10.20.[VLAN_ID].[DEVICE]. DHCP covers IPs .50 to .250, leaving headroom for up to 49 static addresses on each network if necessary. Roughly 200 dynamically allocated addresses should be more than sufficient.

VLAN	SUBNET	ADDRESS	DHCP	USABLE	SPECIFIC HOSTS
2	/24	10.20.2.0	N/A	.1 - .254	.254 ASA inside interface
10	/24	10.20.10.0	.50 - .250	.2 - .254	.10 Printer
20	/24	10.20.20.0	.50 - .250	.2 - .254	.10 Printer
30	/24	10.20.30.0	.50 - .250	.2 - .254	
40	/24	10.20.40.0	.50 - .250	.2 - .254	.10 Printer
45	/24	10.20.45.0	.50 - .250	.2 - .254	.10 Printer
50	/24	10.20.50.0	.50 - .250	.2 - .254	
60	/24	10.20.60.0	Static only	.2 - .254	.10 HTTPS web server
70	/24	10.20.70.0	Static only	.2 - .254	.10 DNS and DHCP, .11 AAA and RADIUS, .12 Syslog and NTP
75	/24	10.20.75.0	Static only	.2 - .254	.10 Compute server
80	/24	10.20.80.0	Static only	.2 - .254	.10 FTP storage server
99	/24	10.20.99.0	Static only	.2 - .254	.11–.30 Physical switch SVIs
110	/24	10.20.110.0	.50 - .250	.2 - .254	

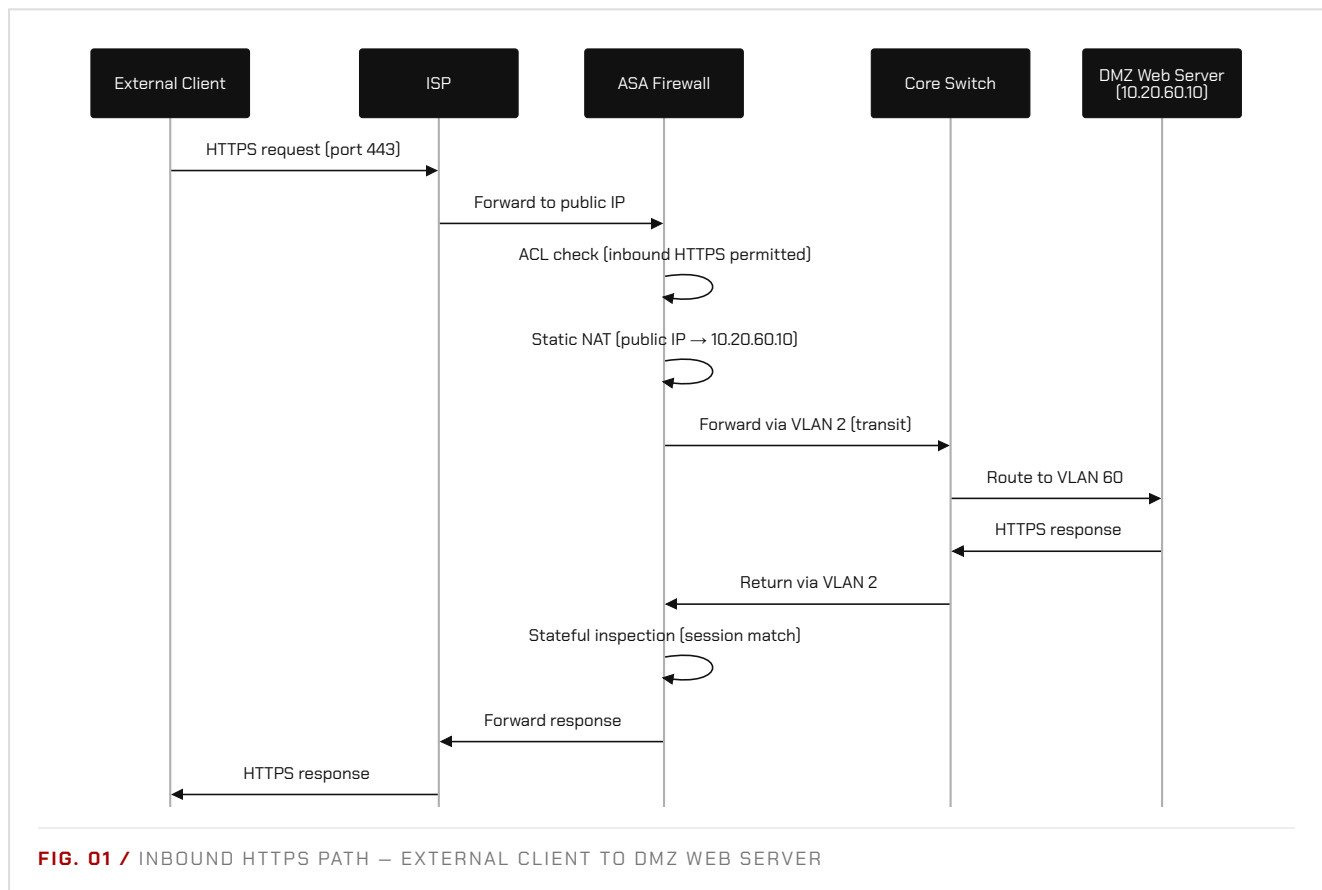
ROUTING DESIGN

7.1 PATHS

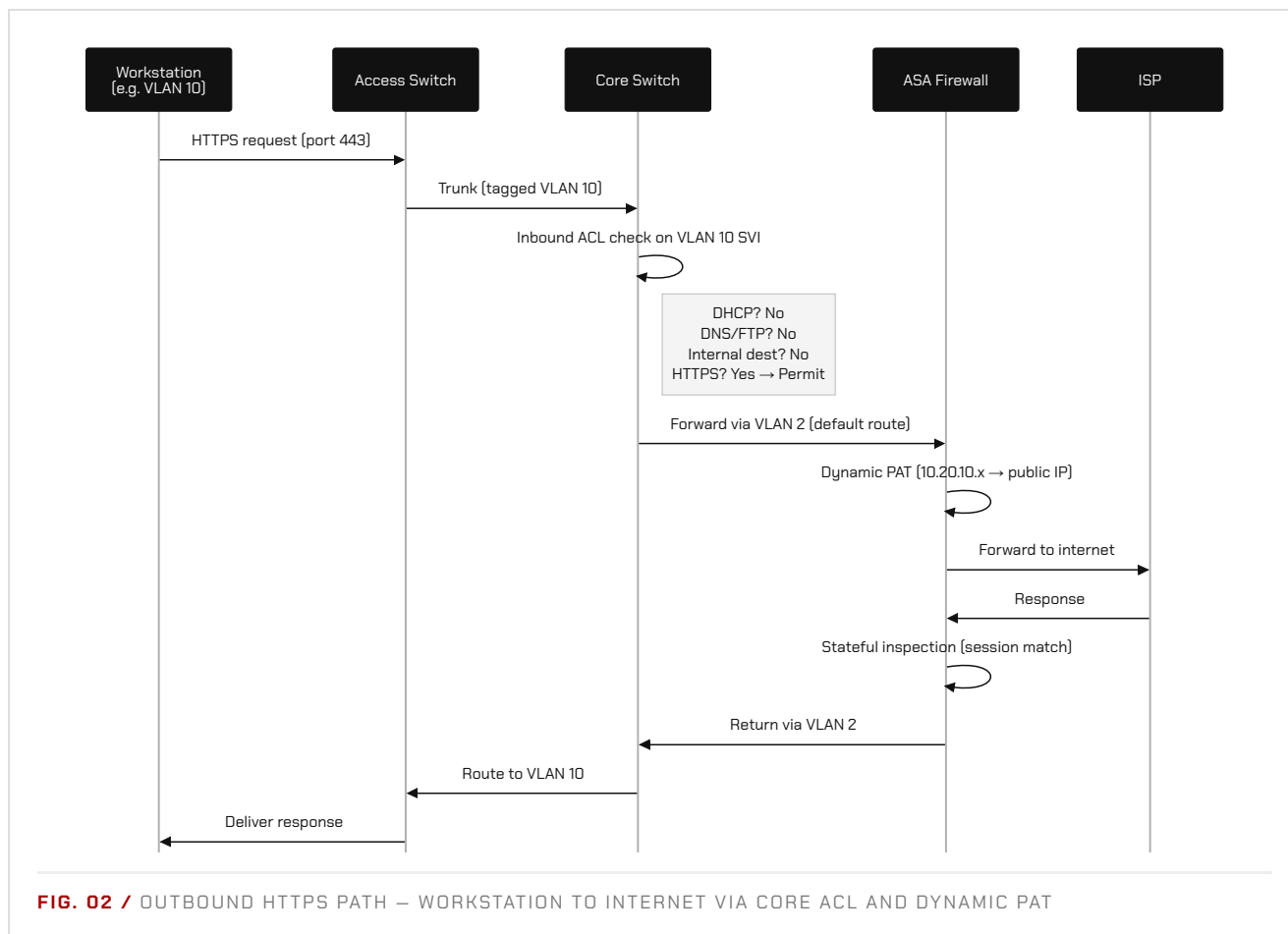
For our purposes, there are three types of routes which we must account for. On the core switches, each VLAN is assigned an ACL group named after itself (such as ACL_VLAN10_IN) which contains the routing rules for inter-VLAN communication. The ASA has its own inbound and outbound ACLs, which we discuss further in this document.

Internal: Packets of this type never leave our network and request services or resources within it. These packets only pass through the core switches and are subject to inbound ACL rules on the core switches.

Inbound: Inbound packets are exclusively meant to access the web interface hosted on the DMZ server. Only HTTPS packets are accepted; they pass through the ASA (which applies its own ACL and NAT rules) and are forwarded to the core switches, which deliver them to the DMZ web server on VLAN 60.



Outbound: Outbound traffic comes from hosts or the DMZ and is routed through the core switches to the ASA, where it is directed to the ISP. Guest WiFi traffic is restricted to HTTPS and DNS and is logged to help identify any misuse.



7.2 HSRPV2

HSRPv2, or "Hot Standby Router Protocol version 2," is a proprietary protocol available on certain Cisco devices such as the core switches in our network. It provides redundancy and ensures that user traffic immediately recovers from hop failures. Both of our core switches share a virtual IP per VLAN, but only one is active at a time; the other serves as a standby in case of failure or to provide easier maintenance. Each VLAN is assigned an HSRP group matching its VLAN ID (e.g., VLAN 10 uses group 10). Preemption is enabled, meaning CORE-1 reclaims the active role automatically upon recovery.

7.3 NAT

Inbound and outbound traffic must pass through the ASA, which contains NAT rules to translate IP addresses. These rules are detailed in the configuration appendices. The NAT provides a path for outbound web traffic via dynamic PAT, as well as a static NAT entry directing inbound HTTPS traffic to the DMZ web server. Stateful inspection permits smooth and secure connection handling at the cost of a somewhat more complex rule set.

SECURITY ARCHITECTURE

8.1 PERIMETER SECURITY

From a networking perspective, the perimeter security is handled exclusively by the ASA. Stateful inspection allows us to keep track of sessions inbound and outbound, gracefully handling legitimate return traffic stemming from outbound connections. This contrasts with the core switches, which only perform stateless inspection via ACLs.

The public-facing (static) IP of our DMZ server is recorded within the ASA's NAT rules, permitting inbound HTTPS traffic to reach the DMZ web server. The ASA also controls which traffic the DMZ server can initiate toward internal resources (such as DNS queries to the internal DNS server).

8.2 DMZ DESIGN

The DMZ is implemented as its own VLAN (VLAN 60) present on both the ASA and the core switches. External traffic reaching the DMZ is handled by the ASA's NAT rules, which direct inbound HTTPS connections to the DMZ web server. The active core switch then delivers the packet to the server.

On the core switches, an inbound ACL on the VLAN 60 SVI restricts what traffic the DMZ server can initiate. Specifically:

- / ICMP echo requests from the DMZ toward internal hosts and external targets are denied, preventing the DMZ from being used to probe or scan the network.
- / ICMP echo-reply is permitted toward the IT subnet only, allowing IT to verify DMZ server connectivity.
- / TCP established traffic is permitted to allow return traffic for legitimate TCP sessions.
- / All other IP traffic from the DMZ to internal subnets (10.20.0.0/16) is denied.
- / Remaining outbound traffic (toward the internet) is permitted.

Traffic from external sources to the DMZ is controlled by the ASA's ACL and NAT policies, not by the core switch ACLs.

While a web server is the only current service on the DMZ, this can be expanded in the future to offer additional services, though this would require a careful update of both the ASA policies and the core switch ACLs.

For demonstration purposes, a mockup website was implemented in the Packet Tracer file itself.

8.3 ACL STRATEGY

Our approach to ACL implementation is based on our access control matrix (see next section). We apply inbound-only ACLs on source VLAN SVIs. Outbound rules would impose additional processing on the core switch before packets can be checked, which decreases global network throughput. Inbound-only filtering is the appropriate approach for a network of this scope where security is a priority.

No ACLs are applied on server VLANs (70, 75, 80). Applying inbound ACLs on these VLANs would block legitimate response traffic (DNS replies, DHCP offers, FTP data) because the core switches perform stateless inspection and cannot distinguish a response from a new connection. Access to these servers is instead controlled at the source: department ACLs determine which VLANs can reach which servers.

The general rule sequence applied to each department VLAN (adapted per department on a least-privilege basis) follows this order:

- 1** Permit DHCP broadcast (required for ip helper-address to function)
- 2** Permit specific internal services (DNS, FTP, compute access, etc., varying by department)
- 3** Permit ICMP echo-reply to the IT subnet (allows IT to verify host connectivity via ping)
- 4** Deny all traffic to internal subnets (10.20.0.0/16, logged), blocking all inter-VLAN traffic not explicitly permitted above
- 5** Permit HTTPS outbound (matches only non-internal destinations since rule 4 has already filtered internal traffic)
- 6** Deny all remaining traffic (logged)

VLAN 99 (Network Management) follows a different pattern. Network devices on this VLAN need to reach specific infrastructure services, so its ACL permits RADIUS (ports 1812/1813) to the AAA server, NTP (port 123) to the NTP server, and syslog (port 514) to the syslog server, then denies everything else. Network devices do not require internet access.

The ACL configuration is generated automatically via a Python script which reads the access control matrix from a spreadsheet and produces paste-ready Cisco IOS commands. This ensures consistency between the documented matrix and the deployed configuration.

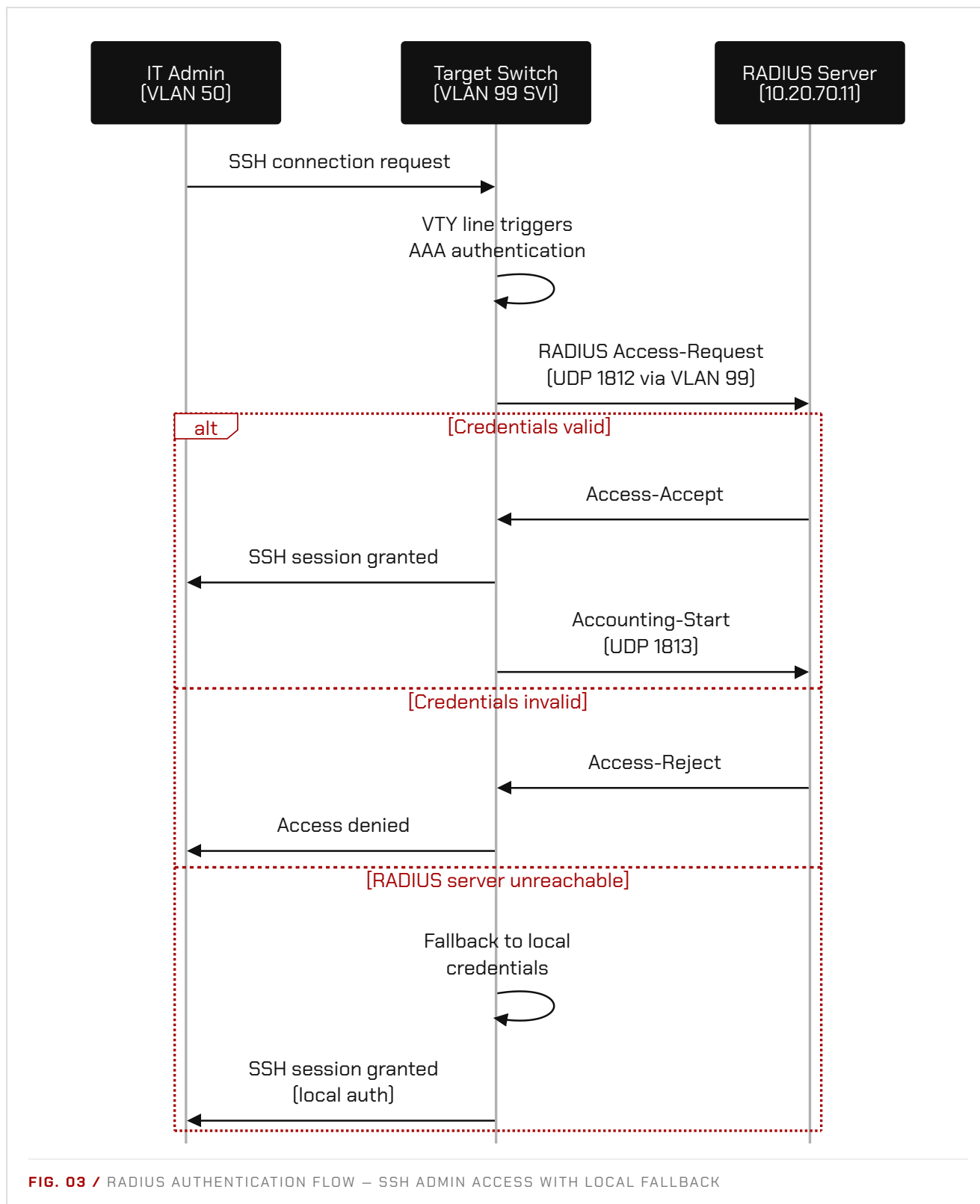
NOTE – A confirmed bug in Cisco Packet Tracer 9.0 prevents ACL bindings on SVI interfaces from taking effect. As a workaround, ACLs are applied to the physical trunk ports on the core switches instead, achieving equivalent filtering. This limitation is specific to the simulator and does not affect production hardware. Full details are documented in the appendix.

8.4 AAA AND RADIUS

We use RADIUS as our authentication protocol, hosted on a server at 10.20.70.11 on VLAN 70. On the RADIUS server, each switch is registered with a corresponding client name, client IP, and passkey. Authentication is enforced on SSH access to all network devices; this traffic passes through VLAN 99 for all devices except for the core switches which employ VLAN 70.

A local user is created on every switch in case the AAA server fails (but remains inaccessible if the AAA server is online), but by default, authentication credentials are provided by the AAA server. Having a centralized AAA server primarily helps IT personnel manage security across multiple devices at once. Each password and username differs between the centralized AAA accounts and the local user accounts, so there is no risk of credential reuse if a threat actor gains access to one device.

Our accounting records denials along with all outbound web traffic coming from the guest WiFi hotspot, along with operations performed by IT.



8.5 DEVICE HARDENING

Each device in the network is configured with security best practices to minimize the margin of maneuver for threat actors. This includes:

- / Telnet connections disabled on all devices
- / SSH version 2 only (SSH version 1 is considered insecure)
- / All passwords encrypted in saved configurations via service password-encryption
- / Legal disclaimer banner displayed on session initiation

- / Strong passwords and unique usernames which do not follow a predictable pattern
- / Console and VTY line timeout delays
- / Strict password attempt limits with lockout on failure
- / All unused ports disabled and assigned to VLAN 999
- / BPDU Guard enabled on host-facing ports of access switches

ACCESS CONTROL MATRIX

The access control matrix provides a high-level reference of the permissions and denials between VLANs. The matrix below does not include every VLAN for the following reasons:

- / **VLAN 2 (Transit):** Governed by ASA security policies, not core switch ACLs.
- / **VLAN 60 (DMZ):** Has its own dedicated ACL on the core switches (see DMZ Design above) and is additionally governed by ASA policies.
- / **VLAN 70 (Internal Servers):** No inbound ACL applied; response traffic must flow freely. Access is controlled at the source VLANs.
- / **VLAN 75 (Compute):** Same rationale as VLAN 70.
- / **VLAN 80 (Storage):** Same rationale as VLAN 70.
- / **VLAN 999 (Blackhole):** No SVI, no routing, no ACL.

SOURCE	DEPT VLANS	SERVERS (70)	COMPUTE (75)	STORAGE (80)	NETMGMT (99)	INTERNET
Management (10)	None	DNS	None	FTP	None	HTTP/S
Study (20)	None	DNS	None	FTP	None	HTTP/S
Production (30)	None	DNS	SSH, Jupyter	FTP	None	HTTP/S
Support A (40)	None	DNS	None	FTP	None	HTTP/S
Support B (45)	None	DNS	None	FTP	None	HTTP/S
IT (50)	SSH	Full	Full	Full	Full	HTTP/S
NetMgmt (99)	None	RADIUS, NTP, Syslog	None	None	N/A	None
Guest (110)	None	None	None	None	None	HTTP/S, DNS

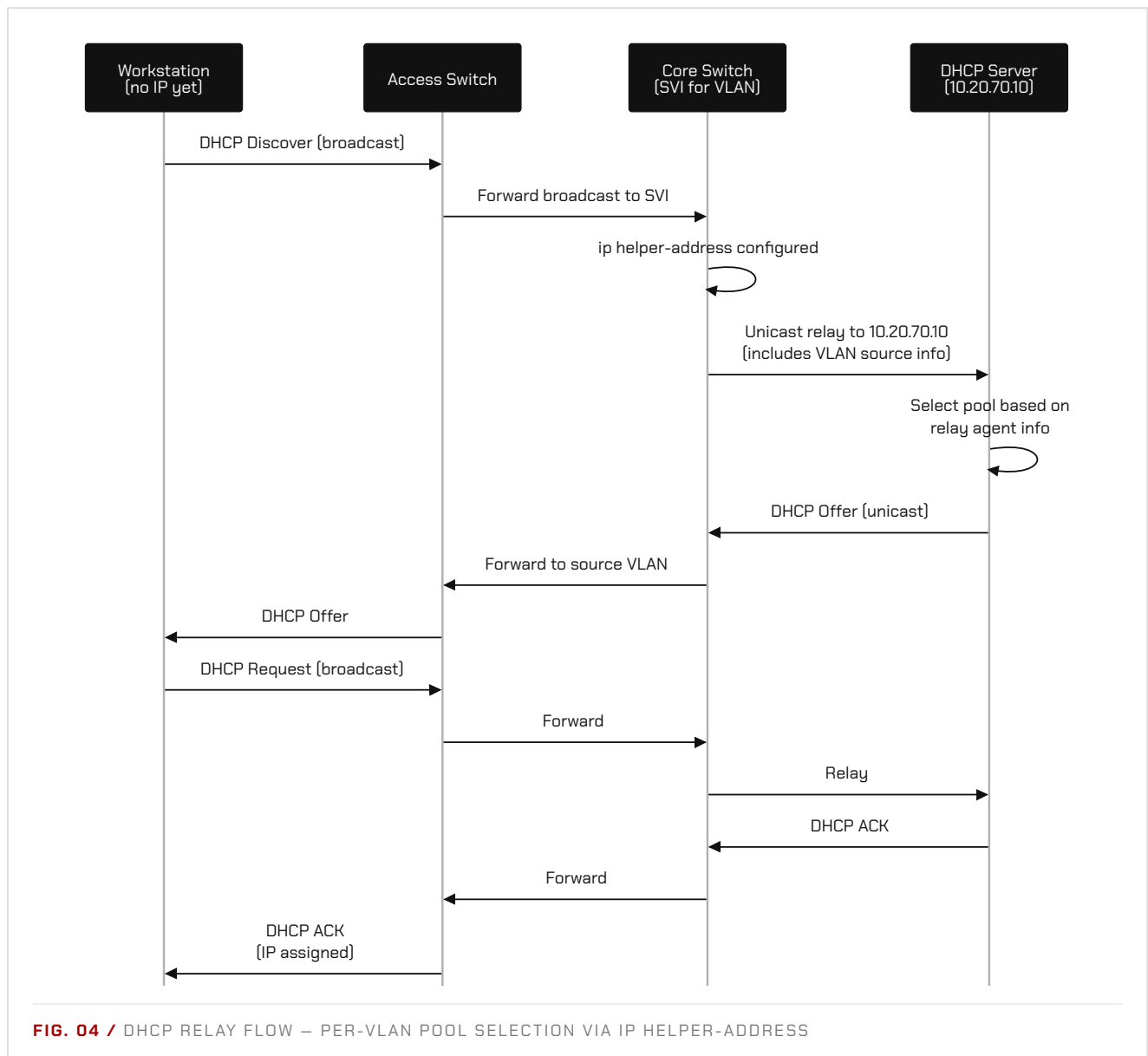
The complete per-VLAN ACL rules are available in **appendix 14.5**.

SERVICES

A series of services are made available to devices on the intranet via servers hosted on VLAN 70 and VLAN 80.

10.1 DHCP

A DHCP server is set up to dynamically attribute IP addresses from pools of approximately 200 addresses available per department (.50 to .250). This does not include printers, which retain static addresses. The pools are relayed to every VLAN via ip helper-address commands configured on each SVI of the core switches.



10.2 DNS

The internal-facing DNS server lets devices on the network refer to NVIDIA's internal services. Thanks to the split-horizon DNS implementation, external users routing through the internet cannot scan for domain names meant to serve only NVIDIA's personnel. All outbound DNS resolution is handled by the internal DNS server, which forwards queries to public resolvers.

10.3 FTP

The FTP server provides a centralized system to store business files, from fiscal records to model weights. Access to various folders can be managed at the application level (read-only or denied) to further containerize data and ensure a continuation of least-privilege principles. The FTP server acts as the sole file-sharing mechanism between departments; inter-department file exchange goes through FTP rather than direct VLAN-to-VLAN transfers.

10.4 HTTPS

A public-facing web server is set up on the DMZ (VLAN 60) to grant the office a web-facing presence. This server can be read and written to by authorized departments but cannot access other hosts and services, as it is isolated within the DMZ. HTTPS certificates are provided by Let's Encrypt.

10.5 NTP

A server provides NTP services to synchronize devices network-wide. This is a key element of our security architecture, as it ensures trustworthy logging timestamps for incident reports and network telemetry, and guarantees that additional security measures such as short-lived encryption certificates can function correctly.

10.6 SYSTEM LOGGING

Logging from the various devices is centralized on the syslog server to record denied traffic attempts, IT operations, and guest WiFi web traffic. User traffic logging is disabled for the time being to extend the duration within which logs can be retained without filling the allocated storage space. While centralizing logs creates a single point of failure, backup practices into FTP or cloud storage (at NVIDIA's discretion) make it a preferable solution to decentralized device-hosted logs, which require considerable overhead in collection and compilation on the part of NVIDIA's IT personnel.

NOTE – A bug in Packet Tracer prevents us from applying logging on the individual ACL rules; we understand this as a limitation of Packet Tracer and the logging would still be implemented on the real application of the network.

CONFIGURATION REFERENCE

This section covers the key steps to configuring the various devices on the network. Complete command configurations can be found in **appendix 14.6**. In the case the network has to be reconfigured from zero, the recommended order in which to bring devices online is:

- 1 CORE-1 and CORE-2
- 2 Services Servers (DHCP, NTP, DNS)
- 3 ACC-SRV (services access switch)
- 4 ASA and DMZ
- 5 All remaining access switches (order not important)

This ensures that the key devices in charge of the system's security are brought online before any device which, if malfunctioning or infected with malware, could have an opportunity to exploit the network.

11.1 CORE SWITCHES

- 1 Basic configuration (hostname, disable DNS lookup, enable routing)
- 2 Set up names for all VLANs
- 3 Set up EtherChannel to bring Po1 to CORE-2
- 4 Configure trunk ports to access switches
- 5 Set up ASA link port
- 6 Configure ACL rules for all VLANs (in Packet Tracer, these are applied to physical trunk ports as a workaround to the SVI ACL binding bug)
- 7 Save

11.2 SERVICES SERVERS

DHCP:

- 1 Set DHCP server IP
- 2 Add pools for CORE-1 and CORE-2 with corresponding interfaces, pool names, default gateway, DNS, address, mask, and maximum users
- 3 Save

NTP:

- 1 Set time and IP
- 2 Save
- 3 NTP must be enabled on the client devices themselves

DNS:

- 1 Set up internal DNS records (IP, name)
- 2 Set up external DNS records (IP, name)

- 3** Configure the IP address of the server responsible for HTTPS services
- 4** Save

11.3 ASA

- 1** Set up identity, hostname, and management access
- 2** Define physical interfaces and security levels
- 3** Define object networks and set up NAT rules
- 4** Configure ACL rules for outbound and inbound traffic
- 5** Set up access-group mappings
- 6** Configure global routing entries
- 7** Configure inspection engine and policy maps
- 8** Disable unused interfaces
- 9** Save

11.4 ACC-DMZ

- 1** Configure identity, secure system logins, and domain name
- 2** Configure global switching, STP, and NTP profiles
- 3** Configure downstream edge port (web server access)
- 4** Disable unused interfaces
- 5** Configure upstream core trunking infrastructure
- 6** Configure out-of-band management interfaces
- 7** Enforce SSH and disable Telnet
- 8** Save

11.5 REMAINING ACCESS SWITCHES (GENERAL TEMPLATE)

- 1** Basic device configuration
- 2** Set up STP and NTP profiles
- 3** Set up required VLAN names (at minimum: own department VLAN, network management, and blackhole)
- 4** Configure trunk switchports
- 5** Define IP addresses (if applicable)
- 6** Harden device
- 7** Save

11.6 PRINTERS

Consult printer user manual here: [bizhub C268i user's guide \(konicaminolta.eu\)](http://bizhub.C268i.user's%20guide.konicaminolta.eu)

11.7 WIRELESS ACCESS POINT

- 1** On port 0, enable port status, set link speed, and enable full duplex (or select "auto")

- 2** On port 1, set SSID, channel limits (default: 6), and coverage range (adjustable, default 140 meters)
- 3** Set authentication method to WPA2-PSK
- 4** Set PSK passphrase
- 5** Set encryption type to AES

VULNERABILITIES AND POINTS OF FAILURE#12

We pride ourselves in creating robust network architectures that go above and beyond our clients' needs, but it would be dishonest not to acknowledge and document the weaknesses of our network, the best practices required to avoid incidents, and the measures needed to protect against threat actors.

12.1 SINGLE POINTS OF FAILURE

This network features several single points of failure, distributed across the network to prevent complete interruptions in any single failure scenario. These include:

- / Single ISP provider
- / Single ASA firewall
- / Single access switch per department
- / Single instance of each service server
- / Single ethernet cable between most devices (with the exception of the EtherChannel link between core switches)

This led to our decision to make the core switches a redundant duo rather than a single unit. We also equip the server room with UPS units capable of sustaining the network long enough to trigger safe shutdown and avoid corrupting data or leaving key parts of the network powerless and vulnerable.

While best practices would dictate duplication or even triplication of each node in the network, this is a necessary sacrifice to remain under budget while delivering a powerful and secure system.

12.2 ACL DRIFT

This failure occurs when a technician forgets to implement a catch-all deny rule at the end of an ACL sequence, potentially allowing any user to send packets to any port on any device in the affected VLAN group. This is a major vulnerability but strictly a result of human error. Our automated ACL generation from the access control matrix mitigates this risk, as the script always appends a catch-all deny rule to each ACL.

12.3 MINIMAL LOGGING

Logging has been kept to a minimum (denials, IT interventions, guest Wi-Fi) to avoid overloading the core switches and filling the syslog server too quickly. Security-first deployments would implement systematic logging of all network traffic, but this requires more capable devices and servers on top of more specialized software suites required to effectively navigate large volumes of logs across months.

Logging of print jobs sent from workstations is possible at NVIDIA's request, but is made partially redundant by physical access to the printers, which in their current configuration do not require identification to execute jobs.

BUDGET OVERVIEW

The following section comprises a comprehensive, itemized version of our budget, covering both CAPEX and OPEX. We propose three different plans of OPEX (basic, standard, premium) based on NVIDIA's requirements; we remain available to explain in detail what each item in the OPEX section covers, and more importantly, what risks or tasks would be taken on by NVIDIA if it chooses to renounce them. All amounts are expressed in EUR.

13.1 CAPEX

WORKSTATIONS & PERIPHERALS

ITEM	QTY	UNIT PRICE	TOTAL
Geekom A5 Mini PC (Ryzen 7 5825U, 16GB, 512, Win11 Pro)	38	433.70	16480.60
MSI MAG Infinite S3 14NVP-2808EU (i7-14700F, RTX 5070 12GB, 32GB DDR5)	10	2173.86	21738.60
AOC 24B3HA2 23.8" FHD IPS 100Hz w/ speakers	38	94.40	3587.20
AOC Q27G42XE 27" QHD 180Hz IPS G-SYNC (Production)	10	140.22	1402.20
Headset USB — Production	10	—	73.04
Keyboard + Mouse combo (standard USB)	48	—	221.74
Headset USB (Management + IT)	10	—	36.52
SUBTOTAL			43539.90

NETWORK DEVICES

ITEM	QTY	UNIT PRICE	TOTAL
Cisco Catalyst 9300-24T-E (core switch)	2	3571.64	7143.28
Cisco Catalyst 9200L-24T-4G-A (access switch)	6	2032.61	12195.66
Cisco C9200CX-12T-2X2G-E (small access switch)	4	1523.91	6095.64
Cisco FPR1010-ASA-K9 (firewall)	1	803.26	803.26
StarTech RK4236BKB (42U server rack)	1	933.70	933.70
Cat6 patch panel 24-port	2	40.22	80.44
TP-Link EAP670 (Wi-Fi 6 AX5400 AP)	1	179.35	179.35
SUBTOTAL			27431.33

CABLING & PHYSICAL NETWORK

ITEM	QTY	UNIT PRICE	TOTAL
Cat5e cable 305m roll	20	16.30	326.00
Cat6 patch cable 2m	10	3.26	32.60
Cat6 patch cable 0.5m	20	1.63	32.60
LC-LC OM3 duplex fiber 1m	4	27.17	108.68
RJ45 keystone jack Cat5e	190	1.50	285.00
Wall faceplate single/double	30	2.17	65.10
Cable tester (RJ45 + fiber)	1	86.96	86.96
1U horizontal cable manager	4	5.43	21.72
Velcro cable ties 100pcs	2	5.43	10.86
Cable labels / label tape	1	10.87	10.87
19" blanking panels 1U	10	2.17	21.70
Cisco USB console cable RJ45	3	10.87	32.61
IEC C13 power cable 2m	15	5.43	81.45
Rack Grounding cable kit	1	32.61	32.61
Shuko-C19 power cable 3m	5	8.50	42.50
C19-C20	5	9.10	45.50
SUBTOTAL			1236.76

SERVER ROOM

ITEM	QTY	UNIT PRICE	TOTAL
Dell PowerEdge R360 (1U, Xeon E-2400, DDR5 ECC)	3	2700.00	8100.00
Synology DiskStation DS923+ (4-bay NAS)	1	832.22	832.22
Seagate IronWolf Pro 28TB ST28000NT000	4	1110.00	4440.00
WD Red SN700 500GB NVMe M.2	2	218.89	437.78
APC Smart-UPS SMT3000RMI2UNC (3kVA, 2U)	3	2666.67	8000.01
APC Basic Rack PDU AP9572 (16A, 13× C13)	2	133.33	266.66
ATEN CS1716i (16-port KVM over IP)	1	444.44	444.44
SUBTOTAL			22521.11

COMPUTE UNIT

ITEM	QTY	UNIT PRICE	TOTAL
Lenovo ThinkSystem SR675 V3 (3U chassis)	1	6736.84	6736.84
NVIDIA H100 PCIe 80GB HBM3	4	27368.42	109473.68
AMD EPYC 9554 (64-core, 3.1GHz)	2	2947.37	5894.74
Samsung M321R8GA0BB0-CQK 64GB DDR5 ECC RDIMM	8	3156.32	25250.56
7.68TB NVMe Gen5 SSD (enterprise)	2	2526.32	5052.64
Precision AC unit min. 5kW (Stulz/Vertiv/Schneider)	1	6315.79	6315.79
Lenovo Premier Support 3yr (24x7x365 NBD on-site)	1	8947.37	8947.37
SUBTOTAL			167671.62

13.2 LABOR

WORKER	DAYS	TOTAL COST (EUR)
Anton Baudoux De Corte — Project Manager	10	7558.14
Alvi Edilkhanov — Architecture Manager	9	7848.84
Jakub Rutyna — Security Engineer	9	6802.33
Sajjad Shahpoor — Security Engineer	8	6046.51
Kien Truong — Network Engineer	7	4883.72
Dji Gapomo — Network Engineer	6	3488.37
SUBTOTAL		36627.91

13.3 TOTAL CAPEX

ITEM	COST
Workstations	43539.90
Network Devices	27431.33
Cabling & Connectivity	1236.76
Servers	22521.11
Compute	167671.62
Labor	36627.91
TOTAL CAPEX	299028.63

At **EUR 299,028.63**, the total capital expenditure remains within the EUR 300,000 budget envelope defined in the RFQ.

13.4 OPEX

RECURRING COSTS

EXPENSE	RATIONALE	YEARLY COST
Hardware Maintenance	15%/yr of hardware value	36000.00
Power Consumption	Server Room + Compute (heavy use)	6200.00
ISP Subscription	Business fiber line + uptime SLA	6500.00
NVIDIA AI Enterprise License	4×H100 License (NVAIE)	16000.00
Ubuntu Pro / Proxmox Support	Hypervisor + OS enterprise	2000.00
Printer Rental	4× Konica Minolta bizhub C268i MAX	5952.00
SUBTOTAL		72652.00

MANAGED SERVICES

SERVICE	SCOPE	PER MONTH	PER YEAR
Network Monitoring & Alerting	24/7 — switches, firewall, cores	800.00	9600.00
Security Management	Firewall, ACL, AAA/RADIUS	600.00	7200.00
Server & OS Management	Patch, Backup, Updates	466.67	5600.00
AI Compute Monitoring	SR675 V3, GPU use, CUDA drivers	933.33	11200.00
Helpdesk & On-Call Support	Tier 1/2, 50 users, on-site SLA 4h	1066.67	12800.00
Documentation & Reporting	Monthly reports, topology and change log	266.67	3200.00
SUBTOTAL			49600.00

13.5 OPEX PLANS

In order to accommodate NVIDIA's needs, we have decided to break down the operating expenses and our support into three plans to leave NVIDIA some flexibility in purchasing.

ITEM	ESSENTIAL	STANDARD	PREMIUM
Hardware maintenance scope	Network + Server Room	+ AI Compute Unit	Full hardware base
Helpdesk support	Tier 2/3 escalation only	Tier 1/2, business hours, 4h SLA	Tier 1/2/3, 24/7, 2h SLA
Reporting	Quarterly	Monthly	Monthly + exec review
Annual OPEX	EUR 44,117.99	EUR 69,269.36	EUR 75,999.56
Annual Managed Services	EUR 39,786.72	EUR 49,600.00	EUR 56,799.96
Total annual recurring	EUR 83,904.71	EUR 118,869.36	EUR 132,799.52
Year 1 total (CAPEX + recurring)	EUR 382,933.34	EUR 417,897.99	EUR 431,828.15
3-Year total investment	EUR 550,742.76	EUR 655,636.71	EUR 697,427.19
Average annual investment	EUR 183,580.92	EUR 218,545.57	EUR 232,475.73

APPENDICES

CONFIDENTIALITY NOTICE – The credential tables in this appendix are provided for evaluation completeness and are intended for restricted distribution.

14.1 SWITCH PASSWORDS

DEVICE NAME	ENABLE SECRET KEY	LOCAL USERNAME	LOCAL USER PASSWORD
ASA	Becode123	ladm_asa	Becode123
CORE-1	EpCk6Ez9np@U	ladm_core1	mt99nxTEdW%m
CORE-2	235shAvtA2j#	ladm_core2	6jgDsV2xHnB@
ACC-MGMT	mCt\$3qYYFa4a	ladm_accmgmt	rSZN4\$8buTC4
ACC-STUDY	Na\$QTQnKFUv4	ladm_accstudy	%6S63PC9ef9q
ACC-PROD	ZjC\$YG3g8f7F	ladm_accprod	e2BjrD39Djz!
ACC-SUPA	7Wsz%TEZy3eX	ladm_accsupa	HHM3!5QFLkZu
ACC-SUPB	Me7QrnEd7#Uh	ladm_accsupb	3KjzJBG%m9cm
ACC-IT	qR#2LQyF9vpq	ladm_accit	9HWJHhEJ!X52
ACC-SRV	3@bBRDVqBqVh	ladm_accsrv	pG!vn3Agfzr9
ACC-DMZ	U5ysqSYpA\$W4	ladm_accdmz	@RkNEjSj249s
ACC-COMPUTE	PvFAuKzw8!XU	ladm_acccompute	BC\$v9h2mAbUA
ACC-GUEST	mR4!vX9zK2wP	ladm_accguest	zR3#vF7!xK9wM2

14.2 RADIUS CLIENTS + PASSWORDS

CLIENT NAME	CLIENT IP	SECRET KEY
CORE-1	10.20.70.2	EpCk6Ez9np@U
CORE-2	10.20.70.3	235shAvtA2j#
ACC-MGMT	10.20.99.11	mcT\$3qYYFa4a
ACC-STUDY	10.20.99.12	Na\$QTQnKFUv4
ACC-PROD	10.20.99.13	ZjC\$YG3g8f7F
ACC-SUPA	10.20.99.14	7wsz%TEZy3eX
ACC-SUPB	10.20.99.15	Me7QrnEd7#Uh
ACC-IT	10.20.99.16	qR#2LQyF9vpq
ACC-SRV	10.20.99.17	3@bBRDVqBqVh
ACC-DMZ	10.20.99.19	U5ysqSYpA\$W4
ACC-COMPUTE	10.20.99.20	PvFAuKzw8!XU
ACC-GUEST	10.20.99.18	mR4!vX9zK2wP

14.3 CORE SWITCH-SPECIFIC COMMANDS

DEVICE	COMMAND
CORE-1	radius server SRV-AAA address ipv4 10.20.70.11 auth-port 1812 key EpCk6Ez9np@U
CORE-2	radius server SRV-AAA address ipv4 10.20.70.11 auth-port 1812 key 235shAvtA2j#

14.4 ACCESS SWITCH-RADIUS MAPPING

DEVICE	COMMAND
ACC-MGMT	Radius-server host 10.20.70.11 auth-port 1812 key mcT\$3qYYFa4a
ACC-STUDY	Radius-server host 10.20.70.11 auth-port 1812 key Na\$QTQnKFUv4
ACC-PROD	Radius-server host 10.20.70.11 auth-port 1812 key ZjC\$YG3g8f7F
ACC-SUPA	Radius-server host 10.20.70.11 auth-port 1812 key 7Wsz%TEZy3eX
ACC-SUPB	Radius-server host 10.20.70.11 auth-port 1812 key Me7QrnEd7#Uh
ACC-IT	Radius-server host 10.20.70.11 auth-port 1812 key qR#2LQyF9vpq
ACC-SRV	Radius-server host 10.20.70.11 auth-port 1812 key 3@bBRDVqBqVh
ACC-DMZ	Radius-server host 10.20.70.11 auth-port 1812 key U5ysqSYpA\$W4
ACC-COMPUTE	Radius-server host 10.20.70.11 auth-port 1812 key PvFAuKzw8!XU
ACC-GUEST	Radius-server host 10.20.70.11 auth-port 1812 key mR4!vX9zK2wP

14.5 ACL

Complete per-VLAN access control list rule set, as generated from the access control matrix — one table per ACL group. Rules sourced from VLAN 99 (network management passthrough) are repeated within each department ACL group.

ACL_VLAN10_IN VLAN 10 – MANAGEMENT / SECRETARIAT

SEQ.	ACTION	PROTO	SOURCE / MASK	DESTINATION / MASK	PORT	LOG	RATIONALE
10	permit	udp	any	255.255.255.255	dst 67 (bootps)	No	DHCP discover broadcast
20	permit	udp	10.20.10.0 0.0.0.255	10.20.70.10	dst 53 (domain)	No	DNS queries (UDP)
30	permit	tcp	10.20.10.0 0.0.0.255	10.20.70.10	dst 53 (domain)	No	DNS queries (TCP)
40	permit	tcp	10.20.10.0 0.0.0.255	10.20.80.10	dst 21 (ftp)	No	FTP control channel
50	permit	tcp	10.20.10.0 0.0.0.255	10.20.80.10	dst >1023 (gt 1023)	No	FTP passive data channel
60	permit	icmp	10.20.10.0 0.0.0.255	10.20.50.0 0.0.0.255	echo- reply	No	Allow ping responses to IT diagnostics
70	permit	udp	10.20.99.0 0.0.0.255	10.20.70.11	dst 1812	No	RADIUS authentication
80	permit	udp	10.20.99.0 0.0.0.255	10.20.70.11	dst 1813	No	RADIUS accounting
90	permit	udp	10.20.99.0 0.0.0.255	10.20.70.12	dst 123	No	NTP time sync
100	permit	udp	10.20.99.0 0.0.0.255	10.20.70.12	dst 514	No	Syslog
110	deny	ip	10.20.10.0 0.0.0.255	10.20.0.0 0.0.255.255	any	Yes	Block remaining internal (east- west) traffic
130	permit	tcp	10.20.10.0 0.0.0.255	any	dst 443	No	General HTTPS internet access
140	deny	ip	any	any	any	Yes	Explicit deny-all (catch-all)

ACL_VLAN20_IN VLAN 20 – STUDY

SEQ.	ACTION	PROTO	SOURCE / MASK	DESTINATION / MASK	PORT	LOG	RATIONALE
10	permit	udp	any	255.255.255.255	dst 67 (bootps)	No	DHCP discover broadcast
20	permit	udp	10.20.20.0 0.0.0.255	10.20.70.10	dst 53 (domain)	No	DNS queries (UDP)
30	permit	tcp	10.20.20.0 0.0.0.255	10.20.70.10	dst 53 (domain)	No	DNS queries (TCP)
40	permit	tcp	10.20.20.0 0.0.0.255	10.20.80.10	dst 21 (ftp)	No	FTP control channel
50	permit	tcp	10.20.20.0 0.0.0.255	10.20.80.10	dst >1023 (gt 1023)	No	FTP passive data channel
60	permit	icmp	10.20.20.0 0.0.0.255	10.20.50.0 0.0.0.255	echo- reply	No	Allow ping replies back to IT/diagnostics VLAN
70	permit	udp	10.20.99.0 0.0.0.255	10.20.70.11	dst 1812	No	RADIUS authentication (mgmt/trunk passthrough)
80	permit	udp	10.20.99.0 0.0.0.255	10.20.70.11	dst 1813	No	RADIUS accounting (mgmt/trunk passthrough)
90	permit	udp	10.20.99.0 0.0.0.255	10.20.70.12	dst 123	No	NTP time sync (mgmt/trunk passthrough)
100	permit	udp	10.20.99.0 0.0.0.255	10.20.70.12	dst 514	No	Syslog (mgmt/trunk passthrough)
110	deny	ip	10.20.20.0 0.0.0.255	10.20.0.0 0.0.255.255	any	Yes	Block remaining internal (east- west) traffic
130	permit	tcp	10.20.20.0 0.0.0.255	any	dst 443	No	General HTTPS internet access
140	deny	ip	any	any	any	Yes	Explicit deny-all (catch-all)

ACL_VLAN30_IN VLAN 30 – PRODUCTION

SEQ.	ACTION	PROTO	SOURCE / MASK	DESTINATION / MASK	PORT	LOG	RATIONALE
10	permit	udp	any	255.255.255.255	dst 67 (bootps)	No	DHCP discover broadcast
20	permit	udp	10.20.30.0 0.0.0.255	10.20.70.10	dst 53 (domain)	No	DNS queries (UDP)
30	permit	tcp	10.20.30.0 0.0.0.255	10.20.70.10	dst 53 (domain)	No	DNS queries (TCP)
40	permit	tcp	10.20.30.0 0.0.0.255	10.20.80.10	dst 21 (ftp)	No	FTP control channel
50	permit	tcp	10.20.30.0 0.0.0.255	10.20.80.10	dst >1023 (gt 1023)	No	FTP passive data channel
60	permit	tcp	10.20.30.0 0.0.0.255	10.20.75.10	dst 22 (ssh)	No	SSH access to host on .75 subnet
70	permit	tcp	10.20.30.0 0.0.0.255	10.20.75.10	dst 8888	No	App/service access on .75 host (custom port)
80	permit	icmp	10.20.30.0 0.0.0.255	10.20.50.0 0.0.0.255	echo- reply	No	Allow ping replies back to IT/diagnostics VLAN
90	permit	udp	10.20.99.0 0.0.0.255	10.20.70.11	dst 1812	No	RADIUS authentication (mgmt/trunk passthrough)
100	permit	udp	10.20.99.0 0.0.0.255	10.20.70.11	dst 1813	No	RADIUS accounting (mgmt/trunk passthrough)
110	permit	udp	10.20.99.0 0.0.0.255	10.20.70.12	dst 123	No	NTP time sync (mgmt/trunk passthrough)
120	permit	udp	10.20.99.0 0.0.0.255	10.20.70.12	dst 514	No	Syslog (mgmt/trunk passthrough)
130	deny	ip	10.20.30.0 0.0.0.255	10.20.0.0 0.0.255.255	any	Yes	Block remaining internal (east-west) traffic
150	permit	tcp	10.20.30.0 0.0.0.255	any	dst 443	No	General HTTPS internet access
160	deny	ip	any	any	any	Yes	Explicit deny-all (catch-all)

ACL_VLAN40_IN VLAN 40 – SUPPORT A

SEQ.	ACTION	PROTO	SOURCE / MASK	DESTINATION / MASK	PORT	LOG	RATIONALE
10	permit	udp	any	255.255.255.255	dst 67 (bootps)	No	DHCP discover broadcast
20	permit	udp	10.20.40.0 0.0.0.255	10.20.70.10	dst 53 (domain)	No	DNS queries (UDP)
30	permit	tcp	10.20.40.0 0.0.0.255	10.20.70.10	dst 53 (domain)	No	DNS queries (TCP)
40	permit	tcp	10.20.40.0 0.0.0.255	10.20.80.10	dst 21 (ftp)	No	FTP control channel
50	permit	tcp	10.20.40.0 0.0.0.255	10.20.80.10	dst >1023 (gt 1023)	No	FTP passive data channel
60	permit	icmp	10.20.40.0 0.0.0.255	10.20.50.0 0.0.0.255	echo- reply	No	Allow ping replies back to IT/diagnostics VLAN
70	permit	udp	10.20.99.0 0.0.0.255	10.20.70.11	dst 1812	No	RADIUS authentication (mgmt/trunk passthrough)
80	permit	udp	10.20.99.0 0.0.0.255	10.20.70.11	dst 1813	No	RADIUS accounting (mgmt/trunk passthrough)
90	permit	udp	10.20.99.0 0.0.0.255	10.20.70.12	dst 123	No	NTP time sync (mgmt/trunk passthrough)
100	permit	udp	10.20.99.0 0.0.0.255	10.20.70.12	dst 514	No	Syslog (mgmt/trunk passthrough)
110	deny	ip	10.20.40.0 0.0.0.255	10.20.0.0 0.0.255.255	any	Yes	Block remaining internal (east- west) traffic
130	permit	tcp	10.20.40.0 0.0.0.255	any	dst 443	No	General HTTPS internet access
140	deny	ip	any	any	any	Yes	Explicit deny-all (catch-all)

ACL_VLAN45_IN VLAN 45 – SUPPORT B

SEQ.	ACTION	PROTO	SOURCE / MASK	DESTINATION / MASK	PORT	LOG	RATIONALE
10	permit	udp	any	255.255.255.255	dst 67 (bootps)	No	DHCP discover broadcast
20	permit	udp	10.20.45.0 0.0.0.255	10.20.70.10	dst 53 (domain)	No	DNS queries (UDP)
30	permit	tcp	10.20.45.0 0.0.0.255	10.20.70.10	dst 53 (domain)	No	DNS queries (TCP)
40	permit	tcp	10.20.45.0 0.0.0.255	10.20.80.10	dst 21 (ftp)	No	FTP control channel
50	permit	tcp	10.20.45.0 0.0.0.255	10.20.80.10	dst >1023 (gt 1023)	No	FTP passive data channel
60	permit	icmp	10.20.45.0 0.0.0.255	10.20.50.0 0.0.0.255	echo- reply	No	Allow ping replies back to IT/diagnostics VLAN
70	permit	udp	10.20.99.0 0.0.0.255	10.20.70.11	dst 1812	No	RADIUS authentication (mgmt/trunk passthrough)
80	permit	udp	10.20.99.0 0.0.0.255	10.20.70.11	dst 1813	No	RADIUS accounting (mgmt/trunk passthrough)
90	permit	udp	10.20.99.0 0.0.0.255	10.20.70.12	dst 123	No	NTP time sync (mgmt/trunk passthrough)
100	permit	udp	10.20.99.0 0.0.0.255	10.20.70.12	dst 514	No	Syslog (mgmt/trunk passthrough)
110	deny	ip	10.20.45.0 0.0.0.255	10.20.0.0 0.0.255.255	any	Yes	Block remaining internal (east- west) traffic
130	permit	tcp	10.20.45.0 0.0.0.255	any	dst 443	No	General HTTPS internet access
140	deny	ip	any	any	any	Yes	Explicit deny-all (catch-all)

ACL_VLAN50_IN VLAN 50 – IT DEPARTMENT (ELEVATED ACCESS)

SEQ.	ACTION	PROTO	SOURCE / MASK	DESTINATION / MASK	PORT	LOG	RATIONALE
10	permit	udp	any	255.255.255.255	dst 67 (bootps)	No	DHCP discover broadcast
20	permit	ip	10.20.50.0 0.0.0.255	10.20.60.0 0.0.0.255	any	No	Full IP access to VLAN60 (likely another infra/server subnet)
30	permit	ip	10.20.50.0 0.0.0.255	10.20.70.0 0.0.0.255	any	No	Full IP access to core/mgmt subnet (DNS/DHCP/AAA/NTP/syslog servers)
40	permit	ip	10.20.50.0 0.0.0.255	10.20.75.0 0.0.0.255	any	No	Full IP access to subnet hosting the SSH/8888 host VLAN30 reached
50	permit	ip	10.20.50.0 0.0.0.255	10.20.80.0 0.0.0.255	any	No	Full IP access to FTP/storage subnet
60	permit	ip	10.20.50.0 0.0.0.255	10.20.99.0 0.0.0.255	any	No	Full IP access to mgmt VLAN
70	permit	tcp	10.20.50.0 0.0.0.255	10.20.10.0 0.0.0.255	dst 22 (ssh)	No	SSH into VLAN10 hosts
80	permit	tcp	10.20.50.0 0.0.0.255	10.20.20.0 0.0.0.255	dst 22 (ssh)	No	SSH into VLAN20 hosts
90	permit	tcp	10.20.50.0 0.0.0.255	10.20.30.0 0.0.0.255	dst 22 (ssh)	No	SSH into VLAN30 hosts
100	permit	tcp	10.20.50.0 0.0.0.255	10.20.40.0 0.0.0.255	dst 22 (ssh)	No	SSH into VLAN40 hosts
110	permit	tcp	10.20.50.0 0.0.0.255	10.20.45.0 0.0.0.255	dst 22 (ssh)	No	SSH into VLAN45 hosts
120	permit	icmp	10.20.50.0 0.0.0.255	10.20.0.0 0.0.255.255	any	No	Ping any internal host (IT diagnostics)
130	deny	ip	10.20.50.0 0.0.0.255	10.20.0.0 0.0.255.255	any	Yes	Block any other internal traffic not explicitly permitted above
150	permit	tcp	10.20.50.0 0.0.0.255	any	dst 443	No	General HTTPS internet access
160	deny	ip	any	any	any	Yes	Explicit deny-all (catch-all)

ACL_VLAN60_IN VLAN 60 – DMZ (DMZ_ISOLATION)

SEQ.	ACTION	PROTO	SOURCE / MASK	DESTINATION / MASK	PORT	LOG	RATIONALE
10	deny	icmp	10.20.60.0 0.0.0.255	10.20.0.0 0.0.255.255	echo (ping request)	Yes	Block DMZ from pinging into the internal network stops the DMZ from being used to probe/scan internal hosts
20	deny	icmp	10.20.60.0 0.0.0.255	any	echo (ping request)	Yes	Block DMZ from pinging anywhere at all, prevents a DMZ host from being used to scan external targets
30	permit	icmp	10.20.60.0 0.0.0.255	10.20.0.0 0.0.255.255	echo- reply	No	Allow the DMZ to reply to pings so internal monitoring/IT can still health-check the server by pinging it from inside
40	permit	tcp	any	any	established	No	Allow return traffic for already-established TCP sessions, so replies to legitimate connections aren't blocked
50	deny	ip	10.20.60.0 0.0.0.255	10.20.0.0 0.0.255.255	any	Yes	Core isolation rule blocks all other DMZ→internal traffic not already explicitly permitted above
60	permit	ip	10.20.60.0 0.0.0.255	any	any	No	Allow everything else outbound from the DMZ (e.g. to the internet)

ACL_VLAN110_IN VLAN 110 – GUEST WIFI

SEQ.	ACTION	PROTO	SOURCE / MASK	DESTINATION / MASK	PORT	LOG	RATIONALE
10	permit	udp	any	255.255.255.255	dst 67 (bootps)	No	DHCP discover broadcast
20	permit	udp	10.20.110.0 0.0.0.255	10.20.70.10	dst 53 (domain)	Yes	DNS queries (UDP)
30	permit	tcp	10.20.110.0 0.0.0.255	10.20.70.10	dst 53 (domain)	Yes	DNS queries (TCP)
40	deny	icmp	10.20.110.0 0.0.0.255	10.20.0.0 0.0.255.255	echo	Yes	Block guests from pinging into the internal network (recon prevention)
50	permit	icmp	10.20.110.0 0.0.0.255	10.20.50.0 0.0.0.255	echo-reply	No	Allow ping replies back to IT/diagnostics only
60	deny	ip	10.20.110.0 0.0.0.255	10.20.0.0 0.0.255.255	any	Yes	Core isolation — block all other guest→internal traffic
70	permit	tcp	10.20.110.0 0.0.0.255	any	dst 443	Yes	Outbound HTTPS (internet access)
80	deny	ip	any	any	any	Yes	Explicit deny-all (catch-all)

14.6 FULL DEVICE CONFIGURATIONS

Paste-ready device configurations. Each frame below contains the complete command sequence for one device; use the COPY ALL button (or triple-click and drag inside a frame) to copy the entire configuration in a single selection.

CORE-1

```
enable
configure terminal
hostname CORE-1
no ip domain-lookup
no logging console
ip routing
! VLANs (14 total incl. default)
vlan 2
    name TRANSIT
vlan 10
    name MGMT_SECRETARIAT
vlan 20
    name STUDY
vlan 30
    name PRODUCTION
vlan 40
    name SUPPORT_A
vlan 45
    name SUPPORT_B
vlan 50
    name IT_DEPT
vlan 60
    name DMZ
vlan 70
    name INT_SERVERS
vlan 75
    name SRV_COMPUTE
vlan 80
    name STORAGE
vlan 99
    name NET_MGMT
vlan 999
    name BLACKHOLE
exit
! EtherChannel (Po1 to CORE-2)
interface range GigabitEthernet1/0/1-2
    switchport mode trunk
    switchport trunk native vlan 999
    channel-group 1 mode active
    no shutdown
! Trunk ports to access switches (Gig1/0/3-10)
interface range GigabitEthernet1/0/3-10
    switchport mode trunk
    switchport trunk native vlan 999
    no shutdown
! ASA link port (Gig1/0/24)
interface GigabitEthernet1/0/24
    switchport mode access
    switchport access vlan 2
    no shutdown
end
! =====
! ACL Configuration – applied to physical trunk ports
! (PT 9.0 SVI bug workaround – see ACL_SVI_Bug_Incident_Report.md)
! =====

! --- VLAN 10 – Management/Secretariat ---
ip access-list extended ACL_VLAN10_IN
10 permit udp any host 255.255.255.255 eq bootps          ! DHCP
20 permit udp 10.20.10.0 0.0.0.255 host 10.20.70.10 eq domain ! DNS UDP
30 permit tcp 10.20.10.0 0.0.0.255 host 10.20.70.10 eq domain ! DNS TCP
40 permit tcp 10.20.10.0 0.0.0.255 host 10.20.80.10 eq ftp    ! FTP control
50 permit tcp 10.20.10.0 0.0.0.255 host 10.20.80.10 gt 1023   ! FTP passive data
60 permit icmp 10.20.10.0 0.0.0.255 10.20.50.0 0.0.0.255 echo-reply ! ICMP replies to IT
70 permit udp 10.20.99.0 0.0.0.255 host 10.20.70.11 eq 1812   ! RADIUS auth
80 permit udp 10.20.99.0 0.0.0.255 host 10.20.70.11 eq 1813   ! RADIUS accounting
```

```

90 permit udp 10.20.99.0 0.0.0.255 host 10.20.70.12 eq 123      ! NTP
100 permit udp 10.20.99.0 0.0.0.255 host 10.20.70.12 eq 514    ! Syslog
110 deny ip 10.20.10.0 0.0.0.255 10.20.0.0 0.0.255.255        ! Block internal
130 permit tcp 10.20.10.0 0.0.0.255 any eq 443                 ! HTTPS
140 deny ip any any                                           ! Deny all

interface gig1/0/3
switchport trunk native vlan 999
switchport mode trunk
ip access-group ACL_VLAN10_IN in

! --- VLAN 20 - Study ---
ip access-list extended ACL_VLAN20_IN
10 permit udp any host 255.255.255.255 eq bootps              ! DHCP
20 permit udp 10.20.20.0 0.0.0.255 host 10.20.70.10 eq domain ! DNS UDP
30 permit tcp 10.20.20.0 0.0.0.255 host 10.20.70.10 eq domain ! DNS TCP
40 permit tcp 10.20.20.0 0.0.0.255 host 10.20.80.10 eq ftp     ! FTP control
50 permit tcp 10.20.20.0 0.0.0.255 host 10.20.80.10 gt 1023    ! FTP passive data
60 permit icmp 10.20.20.0 0.0.0.255 10.20.50.0 0.0.0.255 echo-reply ! ICMP replies to IT
70 permit udp 10.20.99.0 0.0.0.255 host 10.20.70.11 eq 1812   ! RADIUS auth
80 permit udp 10.20.99.0 0.0.0.255 host 10.20.70.11 eq 1813   ! RADIUS accounting
90 permit udp 10.20.99.0 0.0.0.255 host 10.20.70.12 eq 123     ! NTP
100 permit udp 10.20.99.0 0.0.0.255 host 10.20.70.12 eq 514    ! Syslog
110 deny ip 10.20.20.0 0.0.0.255 10.20.0.0 0.0.255.255        ! Block internal
130 permit tcp 10.20.20.0 0.0.0.255 any eq 443                 ! HTTPS
140 deny ip any any                                           ! Deny all

interface gig1/0/4
switchport trunk native vlan 999
switchport mode trunk
ip access-group ACL_VLAN20_IN in

! --- VLAN 30 - Production (extra: SSH + Jupyter to compute server) ---
ip access-list extended ACL_VLAN30_IN
10 permit udp any host 255.255.255.255 eq bootps              ! DHCP
20 permit udp 10.20.30.0 0.0.0.255 host 10.20.70.10 eq domain ! DNS UDP
30 permit tcp 10.20.30.0 0.0.0.255 host 10.20.70.10 eq domain ! DNS TCP
40 permit tcp 10.20.30.0 0.0.0.255 host 10.20.80.10 eq ftp     ! FTP control
50 permit tcp 10.20.30.0 0.0.0.255 host 10.20.80.10 gt 1023    ! FTP passive data
60 permit tcp 10.20.30.0 0.0.0.255 host 10.20.75.10 eq 22      ! SSH to compute
70 permit tcp 10.20.30.0 0.0.0.255 host 10.20.75.10 eq 8888    ! Jupyter notebook
80 permit icmp 10.20.30.0 0.0.0.255 10.20.50.0 0.0.0.255 echo-reply ! ICMP replies to IT
90 permit udp 10.20.99.0 0.0.0.255 host 10.20.70.11 eq 1812   ! RADIUS auth
100 permit udp 10.20.99.0 0.0.0.255 host 10.20.70.11 eq 1813  ! RADIUS accounting
110 permit udp 10.20.99.0 0.0.0.255 host 10.20.70.12 eq 123    ! NTP
120 permit udp 10.20.99.0 0.0.0.255 host 10.20.70.12 eq 514    ! Syslog
130 deny ip 10.20.30.0 0.0.0.255 10.20.0.0 0.0.255.255        ! Block internal
150 permit tcp 10.20.30.0 0.0.0.255 any eq 443                 ! HTTPS
160 deny ip any any                                           ! Deny all

interface gig1/0/5
switchport trunk native vlan 999
switchport mode trunk
ip access-group ACL_VLAN30_IN in

! --- VLAN 40 - Support A ---
ip access-list extended ACL_VLAN40_IN
10 permit udp any host 255.255.255.255 eq bootps              ! DHCP
20 permit udp 10.20.40.0 0.0.0.255 host 10.20.70.10 eq domain ! DNS UDP
30 permit tcp 10.20.40.0 0.0.0.255 host 10.20.70.10 eq domain ! DNS TCP
40 permit tcp 10.20.40.0 0.0.0.255 host 10.20.80.10 eq ftp     ! FTP control
50 permit tcp 10.20.40.0 0.0.0.255 host 10.20.80.10 gt 1023    ! FTP passive data
60 permit icmp 10.20.40.0 0.0.0.255 10.20.50.0 0.0.0.255 echo-reply ! ICMP replies to IT
70 permit udp 10.20.99.0 0.0.0.255 host 10.20.70.11 eq 1812   ! RADIUS auth
80 permit udp 10.20.99.0 0.0.0.255 host 10.20.70.11 eq 1813   ! RADIUS accounting
90 permit udp 10.20.99.0 0.0.0.255 host 10.20.70.12 eq 123     ! NTP
100 permit udp 10.20.99.0 0.0.0.255 host 10.20.70.12 eq 514    ! Syslog
110 deny ip 10.20.40.0 0.0.0.255 10.20.0.0 0.0.255.255        ! Block internal
130 permit tcp 10.20.40.0 0.0.0.255 any eq 443                 ! HTTPS
140 deny ip any any                                           ! Deny all

interface gig1/0/6
switchport trunk native vlan 999
switchport mode trunk
ip access-group ACL_VLAN40_IN in

! --- VLAN 45 - Support B ---
ip access-list extended ACL_VLAN45_IN
10 permit udp any host 255.255.255.255 eq bootps              ! DHCP
20 permit udp 10.20.45.0 0.0.0.255 host 10.20.70.10 eq domain ! DNS UDP
30 permit tcp 10.20.45.0 0.0.0.255 host 10.20.70.10 eq domain ! DNS TCP
40 permit tcp 10.20.45.0 0.0.0.255 host 10.20.80.10 eq ftp     ! FTP control

```

```

50 permit tcp 10.20.45.0 0.0.0.255 host 10.20.80.10 gt 1023      ! FTP passive data
60 permit icmp 10.20.45.0 0.0.0.255 10.20.50.0 0.0.0.255 echo-reply ! ICMP replies to IT
70 permit udp 10.20.99.0 0.0.0.255 host 10.20.70.11 eq 1812    ! RADIUS auth
80 permit udp 10.20.99.0 0.0.0.255 host 10.20.70.11 eq 1813    ! RADIUS accounting
90 permit udp 10.20.99.0 0.0.0.255 host 10.20.70.12 eq 123     ! NTP
100 permit udp 10.20.99.0 0.0.0.255 host 10.20.70.12 eq 514    ! Syslog
110 deny ip 10.20.45.0 0.0.0.255 10.20.0.0 0.0.255.255        ! Block internal
130 permit tcp 10.20.45.0 0.0.0.255 any eq 443                 ! HTTPS
140 deny ip any any                                           ! Deny all

interface gig1/0/7
switchport trunk native vlan 999
switchport mode trunk
ip access-group ACL_VLAN45_IN in

! --- VLAN 50 - IT Department (elevated access) ---
ip access-list extended ACL_VLAN50_IN
10 permit udp any host 255.255.255.255 eq bootps              ! DHCP
20 permit ip 10.20.50.0 0.0.0.255 10.20.60.0 0.0.0.255       ! Full access DMZ
30 permit ip 10.20.50.0 0.0.0.255 10.20.70.0 0.0.0.255       ! Full access internal servers
40 permit ip 10.20.50.0 0.0.0.255 10.20.75.0 0.0.0.255       ! Full access compute
50 permit ip 10.20.50.0 0.0.0.255 10.20.80.0 0.0.0.255       ! Full access storage
60 permit ip 10.20.50.0 0.0.0.255 10.20.99.0 0.0.0.255       ! Full access MGMT VLAN
70 permit tcp 10.20.50.0 0.0.0.255 10.20.10.0 0.0.0.255 eq 22 ! SSH to MGMT
80 permit tcp 10.20.50.0 0.0.0.255 10.20.20.0 0.0.0.255 eq 22 ! SSH to Study
90 permit tcp 10.20.50.0 0.0.0.255 10.20.30.0 0.0.0.255 eq 22 ! SSH to Production
100 permit tcp 10.20.50.0 0.0.0.255 10.20.40.0 0.0.0.255 eq 22 ! SSH to Support A
110 permit tcp 10.20.50.0 0.0.0.255 10.20.45.0 0.0.0.255 eq 22 ! SSH to Support B
120 permit icmp 10.20.50.0 0.0.0.255 10.20.0.0 0.0.255.255   ! ICMP to all internal
130 deny ip 10.20.50.0 0.0.0.255 10.20.0.0 0.0.255.255       ! Block remaining internal
150 permit tcp 10.20.50.0 0.0.0.255 any eq 443                 ! HTTPS
160 deny ip any any                                           ! Deny all

interface gig1/0/8
switchport trunk native vlan 999
switchport mode trunk
ip access-group ACL_VLAN50_IN in

! --- VLAN 60 - DMZ ---
ip access-list extended DMZ_ISOLATION
10 deny icmp 10.20.60.0 0.0.0.255 10.20.0.0 0.0.255.255 echo
20 deny icmp 10.20.60.0 0.0.0.255 any echo
30 permit icmp 10.20.60.0 0.0.0.255 10.20.0.0 0.0.255.255 echo-reply
40 permit tcp any any established
50 deny ip 10.20.60.0 0.0.0.255 10.20.0.0 0.0.255.255
60 permit ip 10.20.60.0 0.0.0.255 any

interface gig1/0/10
switchport trunk native vlan 999
switchport mode trunk
ip access-group DMZ_ISOLATION in
Exit

login block-for 120 attempt 5 within 60
exit
write memory

```

CORE-2

```

enable
configure terminal
hostname CORE-2
no ip domain-lookup
no logging console
ip routing
! VLANs (14 total incl. default)
vlan 2
name TRANSIT
vlan 10
name MGMT_SECRETARIAT
vlan 20
name STUDY
vlan 30
name PRODUCTION
vlan 40
name SUPPORT_A
vlan 45

```

```

name SUPPORT_B
vlan 50
name IT_DEPT
vlan 60
name DMZ
vlan 70
name INT_SERVERS
vlan 75
name SRV_COMPUTE
vlan 80
name STORAGE
vlan 99
name NET_MGMT
vlan 999
name BLACKHOLE
exit
! EtherChannel (Po1 to CORE-1) – no ASA port on CORE-2
interface range GigabitEthernet1/0/1-2
switchport mode trunk
switchport trunk native vlan 999
channel-group 1 mode active
no shutdown
! Trunk ports to access switches (Gig1/0/3-10)
interface range GigabitEthernet1/0/3-10
switchport mode trunk
switchport trunk native vlan 999
no shutdown
end
write memory
! =====
! ACL Configuration – applied to physical trunk ports
! (PT 9.0 SVI bug workaround – see ACL_SVI_Bug_Incident_Report.md)
! =====

! --- VLAN 10 – Management/Secretariat ---
ip access-list extended ACL_VLAN10_IN
10 permit udp any host 255.255.255.255 eq bootps          ! DHCP
20 permit udp 10.20.10.0 0.0.0.255 host 10.20.70.10 eq domain ! DNS UDP
30 permit tcp 10.20.10.0 0.0.0.255 host 10.20.70.10 eq domain ! DNS TCP
40 permit tcp 10.20.10.0 0.0.0.255 host 10.20.80.10 eq ftp    ! FTP control
50 permit tcp 10.20.10.0 0.0.0.255 host 10.20.80.10 gt 1023   ! FTP passive data
60 permit icmp 10.20.10.0 0.0.0.255 10.20.50.0 0.0.0.255 echo-reply ! ICMP replies to IT
70 permit udp 10.20.99.0 0.0.0.255 host 10.20.70.11 eq 1812   ! RADIUS auth
80 permit udp 10.20.99.0 0.0.0.255 host 10.20.70.11 eq 1813   ! RADIUS accounting
90 permit udp 10.20.99.0 0.0.0.255 host 10.20.70.12 eq 123    ! NTP
100 permit udp 10.20.99.0 0.0.0.255 host 10.20.70.12 eq 514   ! Syslog
110 deny ip 10.20.10.0 0.0.0.255 10.20.0.0 0.0.255.255       ! Block internal
130 permit tcp 10.20.10.0 0.0.0.255 any eq 443                ! HTTPS
140 deny ip any any                                           ! Deny all
interface gig1/0/3
switchport trunk native vlan 999
switchport mode trunk
ip access-group ACL_VLAN10_IN in

! --- VLAN 20 – Study ---
ip access-list extended ACL_VLAN20_IN
10 permit udp any host 255.255.255.255 eq bootps          ! DHCP
20 permit udp 10.20.20.0 0.0.0.255 host 10.20.70.10 eq domain ! DNS UDP
30 permit tcp 10.20.20.0 0.0.0.255 host 10.20.70.10 eq domain ! DNS TCP
40 permit tcp 10.20.20.0 0.0.0.255 host 10.20.80.10 eq ftp    ! FTP control
50 permit tcp 10.20.20.0 0.0.0.255 host 10.20.80.10 gt 1023   ! FTP passive data
60 permit icmp 10.20.20.0 0.0.0.255 10.20.50.0 0.0.0.255 echo-reply ! ICMP replies to IT
70 permit udp 10.20.99.0 0.0.0.255 host 10.20.70.11 eq 1812   ! RADIUS auth
80 permit udp 10.20.99.0 0.0.0.255 host 10.20.70.11 eq 1813   ! RADIUS accounting
90 permit udp 10.20.99.0 0.0.0.255 host 10.20.70.12 eq 123    ! NTP
100 permit udp 10.20.99.0 0.0.0.255 host 10.20.70.12 eq 514   ! Syslog
110 deny ip 10.20.20.0 0.0.0.255 10.20.0.0 0.0.255.255       ! Block internal
130 permit tcp 10.20.20.0 0.0.0.255 any eq 443                ! HTTPS
140 deny ip any any                                           ! Deny all
interface gig1/0/4
switchport trunk native vlan 999
switchport mode trunk
ip access-group ACL_VLAN20_IN in

! --- VLAN 30 – Production (extra: SSH + Jupyter to compute server) ---
ip access-list extended ACL_VLAN30_IN
10 permit udp any host 255.255.255.255 eq bootps          ! DHCP
20 permit udp 10.20.30.0 0.0.0.255 host 10.20.70.10 eq domain ! DNS UDP

```

```

30 permit tcp 10.20.30.0 0.0.0.255 host 10.20.70.10 eq domain ! DNS TCP
40 permit tcp 10.20.30.0 0.0.0.255 host 10.20.80.10 eq ftp ! FTP control
50 permit tcp 10.20.30.0 0.0.0.255 host 10.20.80.10 gt 1023 ! FTP passive data
60 permit tcp 10.20.30.0 0.0.0.255 host 10.20.75.10 eq 22 ! SSH to compute
70 permit tcp 10.20.30.0 0.0.0.255 host 10.20.75.10 eq 8888 ! Jupyter notebook
80 permit icmp 10.20.30.0 0.0.0.255 10.20.50.0 0.0.0.255 echo-reply ! ICMP replies to IT
90 permit udp 10.20.99.0 0.0.0.255 host 10.20.70.11 eq 1812 ! RADIUS auth
100 permit udp 10.20.99.0 0.0.0.255 host 10.20.70.11 eq 1813 ! RADIUS accounting
110 permit udp 10.20.99.0 0.0.0.255 host 10.20.70.12 eq 123 ! NTP
120 permit udp 10.20.99.0 0.0.0.255 host 10.20.70.12 eq 514 ! Syslog
130 deny ip 10.20.30.0 0.0.0.255 10.20.0.0 0.0.255.255 ! Block internal
150 permit tcp 10.20.30.0 0.0.0.255 any eq 443 ! HTTPS
160 deny ip any any ! Deny all
interface gig1/0/5
switchport trunk native vlan 999
switchport mode trunk
ip access-group ACL_VLAN30_IN in

! --- VLAN 40 -- Support A ---
ip access-list extended ACL_VLAN40_IN
10 permit udp any host 255.255.255.255 eq bootps ! DHCP
20 permit udp 10.20.40.0 0.0.0.255 host 10.20.70.10 eq domain ! DNS UDP
30 permit tcp 10.20.40.0 0.0.0.255 host 10.20.70.10 eq domain ! DNS TCP
40 permit tcp 10.20.40.0 0.0.0.255 host 10.20.80.10 eq ftp ! FTP control
50 permit tcp 10.20.40.0 0.0.0.255 host 10.20.80.10 gt 1023 ! FTP passive data
60 permit icmp 10.20.40.0 0.0.0.255 10.20.50.0 0.0.0.255 echo-reply ! ICMP replies to IT
70 permit udp 10.20.99.0 0.0.0.255 host 10.20.70.11 eq 1812 ! RADIUS auth
80 permit udp 10.20.99.0 0.0.0.255 host 10.20.70.11 eq 1813 ! RADIUS accounting
90 permit udp 10.20.99.0 0.0.0.255 host 10.20.70.12 eq 123 ! NTP
100 permit udp 10.20.99.0 0.0.0.255 host 10.20.70.12 eq 514 ! Syslog
110 deny ip 10.20.40.0 0.0.0.255 10.20.0.0 0.0.255.255 ! Block internal
130 permit tcp 10.20.40.0 0.0.0.255 any eq 443 ! HTTPS
140 deny ip any any ! Deny all
interface gig1/0/6
switchport trunk native vlan 999
switchport mode trunk
ip access-group ACL_VLAN40_IN in

! --- VLAN 45 -- Support B ---
ip access-list extended ACL_VLAN45_IN
10 permit udp any host 255.255.255.255 eq bootps ! DHCP
20 permit udp 10.20.45.0 0.0.0.255 host 10.20.70.10 eq domain ! DNS UDP
30 permit tcp 10.20.45.0 0.0.0.255 host 10.20.70.10 eq domain ! DNS TCP
40 permit tcp 10.20.45.0 0.0.0.255 host 10.20.80.10 eq ftp ! FTP control
50 permit tcp 10.20.45.0 0.0.0.255 host 10.20.80.10 gt 1023 ! FTP passive data
60 permit icmp 10.20.45.0 0.0.0.255 10.20.50.0 0.0.0.255 echo-reply ! ICMP replies to IT
70 permit udp 10.20.99.0 0.0.0.255 host 10.20.70.11 eq 1812 ! RADIUS auth
80 permit udp 10.20.99.0 0.0.0.255 host 10.20.70.11 eq 1813 ! RADIUS accounting
90 permit udp 10.20.99.0 0.0.0.255 host 10.20.70.12 eq 123 ! NTP
100 permit udp 10.20.99.0 0.0.0.255 host 10.20.70.12 eq 514 ! Syslog
110 deny ip 10.20.45.0 0.0.0.255 10.20.0.0 0.0.255.255 ! Block internal
130 permit tcp 10.20.45.0 0.0.0.255 any eq 443 ! HTTPS
140 deny ip any any ! Deny all
interface gig1/0/7
switchport trunk native vlan 999
switchport mode trunk
ip access-group ACL_VLAN45_IN in

! --- VLAN 50 -- IT Department (elevated access) ---
ip access-list extended ACL_VLAN50_IN
10 permit udp any host 255.255.255.255 eq bootps ! DHCP
20 permit ip 10.20.50.0 0.0.0.255 10.20.60.0 0.0.0.255 ! Full access DMZ
30 permit ip 10.20.50.0 0.0.0.255 10.20.70.0 0.0.0.255 ! Full access internal servers
40 permit ip 10.20.50.0 0.0.0.255 10.20.75.0 0.0.0.255 ! Full access compute
50 permit ip 10.20.50.0 0.0.0.255 10.20.80.0 0.0.0.255 ! Full access storage
60 permit ip 10.20.50.0 0.0.0.255 10.20.99.0 0.0.0.255 ! Full access MGMT VLAN
70 permit tcp 10.20.50.0 0.0.0.255 10.20.10.0 0.0.0.255 eq 22 ! SSH to MGMT
80 permit tcp 10.20.50.0 0.0.0.255 10.20.20.0 0.0.0.255 eq 22 ! SSH to Study
90 permit tcp 10.20.50.0 0.0.0.255 10.20.30.0 0.0.0.255 eq 22 ! SSH to Production
100 permit tcp 10.20.50.0 0.0.0.255 10.20.40.0 0.0.0.255 eq 22 ! SSH to Support A
110 permit tcp 10.20.50.0 0.0.0.255 10.20.45.0 0.0.0.255 eq 22 ! SSH to Support B
120 permit icmp 10.20.50.0 0.0.0.255 10.20.0.0 0.0.255.255 ! ICMP to all internal
130 deny ip 10.20.50.0 0.0.0.255 10.20.0.0 0.0.255.255 ! Block remaining internal
150 permit tcp 10.20.50.0 0.0.0.255 any eq 443 ! HTTPS
160 deny ip any any ! Deny all
interface gig1/0/8
switchport trunk native vlan 999
switchport mode trunk

```

```

ip access-group ACL_VLAN50_IN in

! --- VLAN 60 - DMZ ---
ip access-list extended DMZ_ISOLATION
  10 deny icmp 10.20.60.0 0.0.0.255 10.20.0.0 0.0.255.255 echo
  20 deny icmp 10.20.60.0 0.0.0.255 any echo
  30 permit icmp 10.20.60.0 0.0.0.255 10.20.0.0 0.0.255.255 echo-reply
  40 permit tcp any any established
  50 deny ip 10.20.60.0 0.0.0.255 10.20.0.0 0.0.255.255
  60 permit ip 10.20.60.0 0.0.0.255 any
interface gig1/0/10
  switchport trunk native vlan 999
  switchport mode trunk
  ip access-group DMZ_ISOLATION in
Exit

login block-for 120 attempt 5 within 60
exit
write memory

```

ASA — PERIMETER-FW

```

enable
configure terminal

! =====
! 1. SYSTEM IDENTITY, HOSTNAME & MANAGEMENT ACCESS
! =====
hostname PERIMETER-FW
domain-name nvidia-rnd.local
enable password Becode123

aaa authentication ssh console LOCAL
username ladm_asa password Becode123
!
ssh 10.20.50.0 255.255.255.0 inside
ssh 10.20.10.0 255.255.255.0 inside
ssh timeout 5
!
telnet timeout 5

=====
! 2. PHYSICAL AND SECURITY INTERFACE DEFINITIONS
! =====
interface GigabitEthernet1/1
  nameif outside
  security-level 0
  ip address 8.8.8.2 255.255.255.0
  no shutdown
exit

interface GigabitEthernet1/2
  nameif inside
  security-level 100
  ip address 10.20.2.254 255.255.255.0
  no shutdown
exit

interface GigabitEthernet1/3
  shutdown
exit

interface GigabitEthernet1/4
  security-level 50
  shutdown
exit

interface GigabitEthernet1/5
  shutdown
exit

interface GigabitEthernet1/6
  shutdown
exit

```

```

interface GigabitEthernet1/7
 shutdown
exit

interface GigabitEthernet1/8
 shutdown
exit

interface Management1/1
 management-only
 shutdown
exit

! =====
! 3. OBJECT NETWORK DEFINITIONS AND NAT RULES
! =====
object network PUBLIC-WEB-IP
 host 8.8.8.10
exit

object network DMZ-WEB-SERVER
 host 10.20.60.10
 nat (inside,outside) static 8.8.8.10
exit

object network INSIDE-NETWORK
 subnet 10.20.0.0 255.255.0.0
 nat (inside,outside) dynamic interface
exit

! =====
! 4. ACCESS CONTROL LISTS (ACLs)
! =====
! --- Access Control Lists (ACLs) ---
! --- Outside Interface Inbound Policies ---

access-list OUTSIDE_IN extended permit tcp any host 8.8.8.10 eq 443
access-list OUTSIDE_IN extended deny icmp any host 8.8.8.10

! Inside inbound
access-list INBOUND_ACL extended permit ip 10.20.60.0 255.255.255.0 10.20.2.0 255.255.255.0
access-list INBOUND_ACL extended permit tcp 10.20.60.0 255.255.255.0 any
access-list INBOUND_ACL extended permit udp 10.20.60.0 255.255.255.0 any eq domain
access-list INBOUND_ACL extended deny ip 10.20.60.0 255.255.255.0 any

access-list INBOUND_ACL extended permit tcp 10.20.0.0 255.255.0.0 any eq www
access-list INBOUND_ACL extended permit tcp 10.20.0.0 255.255.0.0 any eq 443

access-list INBOUND_ACL extended permit udp 10.20.0.0 255.255.0.0 any eq domain

! Scoped ICMP for internal networks only
access-list INBOUND_ACL extended permit icmp 10.20.0.0 255.255.0.0 any echo
access-list INBOUND_ACL extended permit icmp 10.20.0.0 255.255.0.0 any echo-reply

access-list INBOUND_ACL extended permit tcp 10.20.0.0 255.255.0.0 10.20.2.0 255.255.255.0 eq 22
access-list INBOUND_ACL extended deny ip any any

! =====
! 5. SECURITY ACCESS-GROUP MAPS
! =====
access-group OUTSIDE_IN in interface outside
access-group INBOUND_ACL in interface inside

! =====
! 6. GLOBAL ROUTING ENTRIES
! =====
route outside 0.0.0.0 0.0.0.0 8.8.8.1 1
route inside 10.20.0.0 255.255.0.0 10.20.2.1 1
route inside 10.20.60.0 255.255.255.0 10.20.2.1 1

! =====
! 7. INSPECTION ENGINE & POLICY MAPS
! =====
policy-map type inspect dns preset_dns_map
 parameters
  message-length maximum 512
exit

```

```

class-map inspection_default
  match default-inspection-traffic
exit

policy-map global_policy
  class inspection_default
    inspect dns preset_dns_map
    inspect ftp
    inspect h323
    inspect http
    inspect icmp
    inspect tftp
  exit

service-policy global_policy global

! =====
! 8. SAVE CONFIGURATION
! =====
end
write memory

```

ACC-DMZ

```

enable
configure terminal

! =====
! 1. IDENTITY, SECURE SYSTEM LOGINS, AND DOMAIN
! =====
hostname ACC-DMZ
ip domain-name nvidia-rnd.local
ip ssh version 2

enable secret 5 $1$mERr$aenhqOsc72bCtwvWtffog0
username ladm_accdmz secret 5 $1$mERr$bldkvBKLmw2GluCnJZANz/

! --- AAA Security Framework (RADIUS + Local Backup) ---
aaa new-model
aaa authentication login default group radius local
aaa authentication enable default group radius local

radius-server host 10.20.70.11 auth-port 1812 key U5ysqSYpA$W4

ip ssh time-out 60
ip ssh authentication-retries 3
service password-encryption

! =====
! 2. GLOBAL SWITCHING, STP, AND NTP PROFILES
! =====
spanning-tree mode pvst
spanning-tree extend system-id

ntp server 10.20.70.12

! =====
! 3. DOWNSTREAM EDGE PORT CONFIGURATION (WEB SERVER ACCESS PORT)
! =====
interface FastEthernet0/1
  switchport access vlan 60
  switchport mode access
  switchport port-security
  switchport port-security mac-address sticky
  switchport port-security mac-address sticky 00D0.5894.1496
  spanning-tree portfast
  spanning-tree bpduguard enable
exit

! =====
! 4. UNUSED INTERFACES HARDENING (SECURITY BEST PRACTICE)
! =====
interface range FastEthernet0/2 - 24
  shutdown

```

```

exit

! =====
! 5. UPSTREAM CORE TRUNKING INFRASTRUCTURE
! =====
interface GigabitEthernet0/1
  switchport trunk native vlan 999
  switchport trunk allowed vlan 60,99
  switchport mode trunk
exit

interface GigabitEthernet0/2
  switchport trunk native vlan 999
  switchport trunk allowed vlan 60,99
  switchport mode trunk
exit

! =====
! 6. SWITCH OUT-OF-BAND MANAGEMENT INTERFACES
! =====
interface Vlan1
  no ip address
  shutdown
exit

interface Vlan99
  ip address 10.20.99.19 255.255.255.0
  no shutdown
exit

ip default-gateway 10.20.99.1

! =====
! 7. VTY LINES SECURITY HARDENING (FORCE SSH ONLY)
! =====
line con 0
exit

line vty 0 15
  transport input ssh
exit

! =====
! 8. EXECUTION COMPLETE & COMMIT CONFIG
! =====
end
write memory

```

ACC-MGMT

```

enable
configure terminal
hostname ACC-MGMT
no ip domain-lookup
no logging console
vlan 10
  name MGMT_SECRETARIAT
exit
vlan 99
  name NET_MGMT
exit
vlan 999
  name BLACKHOLE
exit
interface range GigabitEthernet0/1-2
  switchport mode trunk
  switchport trunk native vlan 999
  switchport trunk allowed vlan 10,99,999
interface range FastEthernet0/1-5
  switchport mode access
  switchport access vlan 10
  switchport port-security
  switchport port-security maximum 1
  switchport port-security violation shutdown
  switchport port-security mac-address sticky

```

```

spanning-tree portfast
spanning-tree bpduguard enable
interface range FastEthernet0/6-23
  switchport mode access
  switchport access vlan 999
shutdown

interface FastEthernet0/24
switchport access vlan 10
switchport mode access
switchport port-security
switchport port-security maximum 1
switchport port-security violation shutdown
switchport port-security mac-address sticky
spanning-tree portfast
spanning-tree bpduguard enable
interface vlan 99
  ip address 10.20.99.11 255.255.255.0
  no shutdown
exit
ip default-gateway 10.20.99.1

ip ssh time-out 60
ip ssh authentication-retries 3
service password-encryption
end
write memory

```

ACC-STUDY

```

enable
configure terminal
hostname ACC-STUDY
no ip domain-lookup
no logging console
!
vlan 20
  name STUDY
exit
vlan 99
  name NET_MGMT
exit
vlan 999
  name BLACKHOLE
exit
!
interface range GigabitEthernet0/1-2
  switchport mode trunk
  switchport trunk native vlan 999
  switchport trunk allowed vlan 20,99,999
!
interface range FastEthernet0/1-8
  switchport mode access
  switchport access vlan 20
  switchport port-security
  switchport port-security maximum 1
  switchport port-security violation shutdown
  switchport port-security mac-address sticky
  spanning-tree portfast
  spanning-tree bpduguard enable
!
interface range FastEthernet0/9-23
  switchport mode access
  switchport access vlan 999
shutdown
!
interface FastEthernet0/24
switchport access vlan 20
switchport mode access
switchport port-security
switchport port-security maximum 1
switchport port-security violation shutdown
switchport port-security mac-address sticky
spanning-tree portfast
spanning-tree bpduguard enable

```

```
!  
  
interface vlan 99  
  ip address 10.20.99.12 255.255.255.0  
  no shutdown  
exit  
ip default-gateway 10.20.99.1  
!  
  
ip ssh time-out 60  
ip ssh authentication-retries 3  
service password-encryption  
  
end  
write memory
```

ACC-PROD

```
enable  
configure terminal  
hostname ACC-PROD  
no ip domain-lookup  
no logging console  
vlan 30  
  name PRODUCTION  
exit  
vlan 99  
  name NET_MGMT  
exit  
vlan 999  
  name BLACKHOLE  
exit  
interface range GigabitEthernet0/1-2  
  switchport mode trunk  
  switchport trunk native vlan 999  
  switchport trunk allowed vlan 30,99,999  
interface range FastEthernet0/1-10  
  switchport mode access  
  switchport access vlan 30  
  switchport port-security  
  switchport port-security maximum 1  
  switchport port-security violation shutdown  
  switchport port-security mac-address sticky  
  spanning-tree portfast  
  spanning-tree bpduguard enable  
interface range FastEthernet0/11-24  
  switchport mode access  
  switchport access vlan 999  
  shutdown  
interface vlan 99  
  ip address 10.20.99.13 255.255.255.0  
  no shutdown  
exit  
ip default-gateway 10.20.99.1  
  
ip ssh time-out 60  
ip ssh authentication-retries 3  
service password-encryption  
end  
write memory
```

ACC-SUPA

```
enable  
configure terminal  
hostname ACC-SUPA  
no ip domain-lookup  
no logging console  
!  
vlan 40  
  name SUPPORT_A  
exit
```

```

vlan 99
  name NET_MGMT
exit
vlan 999
  name BLACKHOLE
exit
!
interface range GigabitEthernet0/1-2
  switchport mode trunk
  switchport trunk native vlan 999
  switchport trunk allowed vlan 40,99,999
!
interface range FastEthernet0/1-10
  switchport mode access
  switchport access vlan 40
  switchport port-security
  switchport port-security maximum 1
  switchport port-security violation shutdown
  switchport port-security mac-address sticky
  spanning-tree portfast
  spanning-tree bpduguard enable
!
interface range FastEthernet0/11-23
  switchport mode access
  switchport access vlan 999
  shutdown
!

interface FastEthernet0/24
  switchport access vlan 40
  switchport mode access
  switchport port-security
  switchport port-security maximum 1
  switchport port-security violation shutdown
  switchport port-security mac-address sticky
  spanning-tree portfast
  spanning-tree bpduguard enable
interface vlan 99

!

ip address 10.20.99.14 255.255.255.0
no shutdown
exit
ip default-gateway 10.20.99.1

ip ssh time-out 60
ip ssh authentication-retries 3
service password-encryption

end
!
write memory

```

ACC-SUPB

```

enable
configure terminal
hostname ACC-SUPB
no ip domain-lookup
no logging console
vlan 45
  name SUPPORT_B
exit
vlan 99
  name NET_MGMT
exit
vlan 999
  name BLACKHOLE
exit
interface range GigabitEthernet0/1-2
  switchport mode trunk
  switchport trunk native vlan 999
  switchport trunk allowed vlan 45,99,999
interface range FastEthernet0/1-10

```

```
switchport mode access
switchport access vlan 45
switchport port-security
switchport port-security maximum 1
switchport port-security violation shutdown
switchport port-security mac-address sticky
spanning-tree portfast
spanning-tree bpduguard enable
interface range FastEthernet0/11-23
switchport mode access
switchport access vlan 999
shutdown

interface FastEthernet0/24
switchport access vlan 45
switchport mode access
switchport port-security
switchport port-security maximum 1
switchport port-security violation shutdown
switchport port-security mac-address sticky
spanning-tree portfast
spanning-tree bpduguard enable
interface vlan 99
ip address 10.20.99.15 255.255.255.0
no shutdown
exit
ip default-gateway 10.20.99.1

ip ssh time-out 60
ip ssh authentication-retries 3
service password-encryption
end
write memory
```

ACC-IT

```
enable
configure terminal
hostname ACC-IT
no ip domain-lookup
no logging console
vlan 50
name IT_DEPT
exit
vlan 99
name NET_MGMT
exit
vlan 999
name BLACKHOLE
exit
interface range GigabitEthernet0/1-2
switchport mode trunk
switchport trunk native vlan 999
switchport trunk allowed vlan 50,99,999
interface range FastEthernet0/1-5
switchport mode access
switchport access vlan 50
switchport port-security
switchport port-security maximum 1
switchport port-security violation shutdown
switchport port-security mac-address sticky
spanning-tree portfast
spanning-tree bpduguard enable
interface range FastEthernet0/6-24
switchport mode access
switchport access vlan 999
shutdown

interface vlan 99
ip address 10.20.99.16 255.255.255.0
no shutdown
exit
ip default-gateway 10.20.99.1

ip ssh time-out 60
ip ssh authentication-retries 3
```

```
service password-encryption
end
write memory
```

ACC-SRV

```
enable
configure terminal
hostname ACC-SRV
no ip domain-lookup
no logging console
!
vlan 70
    name INT_SERVERS
exit
vlan 80
    name STORAGE
exit
vlan 99
    name NET_MGMT
exit
vlan 999
    name BLACKHOLE
exit
!
interface range GigabitEthernet0/1-2
    switchport mode trunk
    switchport trunk native vlan 999
    switchport trunk allowed vlan 70,80,99,999
!
interface FastEthernet0/1
    switchport mode access
    switchport access vlan 70
    switchport port-security
    switchport port-security maximum 1
    switchport port-security violation shutdown
    switchport port-security mac-address sticky
    spanning-tree portfast
    spanning-tree bpduguard enable
!
interface FastEthernet0/2
    switchport mode access
    switchport access vlan 80
    switchport port-security
    switchport port-security maximum 1
    switchport port-security violation shutdown
    switchport port-security mac-address sticky
    spanning-tree portfast
    spanning-tree bpduguard enable
!
interface FastEthernet0/3
    switchport mode access
    switchport access vlan 70
    switchport port-security
    switchport port-security maximum 1
    switchport port-security violation shutdown
    switchport port-security mac-address sticky
    spanning-tree portfast
    spanning-tree bpduguard enable
!
interface FastEthernet0/4
    switchport mode access
    switchport access vlan 70
    switchport port-security
    switchport port-security maximum 1
    switchport port-security violation shutdown
    switchport port-security mac-address sticky
    spanning-tree portfast
    spanning-tree bpduguard enable
!
interface range FastEthernet0/5-24
    switchport mode access
    switchport access vlan 999
    shutdown
!
```

```
interface vlan 99
 ip address 10.20.99.17 255.255.255.0
 no shutdown
 exit
 ip default-gateway 10.20.99.1
 !

 ip ssh time-out 60
 ip ssh authentication-retries 3
 service password-encryption

end
write memory
```

ACC-COMPUTE

```
enable
configure terminal
hostname ACC-COMPUTE
no ip domain-lookup
no logging console
vlan 75
 name SRV_COMPUTE
exit
vlan 99
 name NET_MGMT
exit
vlan 999
 name BLACKHOLE
exit
interface range GigabitEthernet0/1-2
 switchport mode trunk
 switchport trunk native vlan 999
 switchport trunk allowed vlan 75,99,999
interface FastEthernet0/1
 switchport mode access
 switchport access vlan 75
 switchport port-security
 switchport port-security maximum 1
 switchport port-security violation shutdown
 switchport port-security mac-address sticky
 spanning-tree portfast
 spanning-tree bpduguard enable
interface range FastEthernet0/2-24
 switchport mode access
 switchport access vlan 999
 shutdown
interface vlan 99
 ip address 10.20.99.20 255.255.255.0
 no shutdown
 exit
 ip default-gateway 10.20.99.1

 ip ssh time-out 60
 ip ssh authentication-retries 3
 service password-encryption
end
write memory
```

ACC-GUEST

```
enable
configure terminal
hostname ACC-GUEST
no ip domain-lookup

! VLANs
vlan 99
 name NET_MGMT
vlan 110
 name GUEST_WIFI
vlan 999
```

```

name BLACKHOLE

! AP-facing access port
interface FastEthernet0/1
  switchport mode access
  switchport access vlan 110
  spanning-tree portfast
  spanning-tree bpduguard enable
  no shutdown

! Uplink to CORE-1 – explicit, no range
interface GigabitEthernet0/1
  switchport mode trunk
  switchport trunk native vlan 999
  switchport trunk allowed vlan 99,110,999
  no shutdown

! Uplink to CORE-2 – explicit, no range
interface GigabitEthernet0/2
  switchport mode trunk
  switchport trunk native vlan 999
  switchport trunk allowed vlan 99,110,999
  no shutdown

! Unused ports – hardened
interface range FastEthernet0/2-24
  switchport mode access
  switchport access vlan 999
  shutdown

! Management SVI
interface Vlan99
  ip address 10.20.99.18 255.255.255.0
  no shutdown

ip default-gateway 10.20.99.1
spanning-tree mode rapid-pvst
end
write memory

```

CORE-1 & CORE-2 – GUEST UPLINK PORTS

Open the CLI on the Core-1 & Core-2 switches and apply the following commands:

```

enable
configure terminal

! Configure Core Uplink Ports
interface GigabitEthernet1/0/12
  switchport trunk allowed vlan 99,110,999
  switchport mode trunk
  switchport trunk native vlan 999
  switchport trunk allowed vlan add 110
  no shutdown
exit

```